University of Murcia
Computer Science Faculty

BACHELOR THESIS

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Evaluating ProPicker toward Universal Macromolecule Localization in Cryo-ET

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Abstract

Cryogenic electron tomography provides three-dimensional visualizations, called tomograms, of cellular environments in their native state. While these tomograms preserve vital structural context, they are often characterized by low signal-to-noise ratios, anisotropy and high annotation costs. As a result, manual particle picking does not scale, and classical template-based methods can become brittle when particle conformation, local environment or acquisition conditions change. This bachelor thesis investigates the practical value of universal deep learning detectors for macromolecule localization in cryogenic electron tomography. The work focuses on ProPicker, based on a promptable three-dimensional segmentation model, and analyzes how far this family of detectors can behave as a general tool for new particle targets.

Our main contribution is a reproducible experimental workflow that integrates preprocessing, prompt construction, model adaptation, inference and evaluation. The study combines the real data cytoplasmic ribosome case from EMPIAR-10988 with a synthetic thyroglobulin dataset provided by the AIKE laboratory and generated using PolNet. The experiments show that prompt-only inference contains useful signal, achieving an F_1 score of 0.3074, but does not provide balanced performance when the particle target and dataset change. On EMPIAR-10988, fine-tuning increases the score to 0.5714. On the synthetic dataset, the base model almost collapses, whereas supervised adaptation reaches a mean F_1 of 0.891. The incremental study further shows that four training tomograms already produce a competitive regime, and that twelve tomograms with multi-prompt inference provide the best observed result, with an F_1 score of 0.919.

Finally, the bachelor thesis identifies prompt robustness as the main limitation observed. Even after supervised adaptation, certain prompts trigger a collapse in recall, a variability that cannot be explained by distortions, spatial position or simple angular rules alone. Consequently, while ProPicker does not yet qualify as a universal detector in the strictest sense, it provides a practical and adaptable foundation for the development of more general cryogenic electron tomography detection systems.

Resumen extendido (*Extended Abstract*)

La tomografía crioelectrónica (cryo-ET) es una técnica que permite estudiar la organización molecular en su estado nativo. A diferencia de los métodos de partícula aislada que utilizan muestras purificadas y pierden la información del entorno, la tomografía crioelectrónica reconstruye volúmenes 3D, llamados tomogramas, donde las macromoléculas se observan dentro de sus contextos biológicos originales. Esta característica es interesante en procesos celulares donde no se depende solo de la estructura de una proteína aislada, sino también de su posición, de sus interacciones locales y de la organización espacial de los complejos que la rodean.

Sin embargo, esta capacidad conlleva retos computacionales significativos. Los tomogramas presentan una baja relación señal-ruido y una marcada anisotropía debido al efecto del *missing wedge*. Además, la alta densidad de las muestras provoca que las partículas puedan solaparse en entornos saturados. Por todo esto, la localización de partículas es una fase determinante, ya que las coordenadas obtenidas definen la calidad del posterior promediado y análisis estructural.

Problema y objetivo.

Esta investigación aborda la localización automática de macromoléculas mediante detectores basados en DL y condicionados por ejemplos o *prompts*. El objetivo es que, a partir de un tomograma y un ejemplo de la partícula de interés, el sistema genere una lista de coordenadas precisas. La investigación no se limita a evaluar el rendimiento en un único conjunto de datos, sino que analiza la capacidad del modelo para mantener su utilidad al cambiar de dominio, de partícula o de condiciones de adquisición. Esta capacidad de generalización se plantea como una solución práctica para reducir la dependencia de anotaciones extensas y evitar entrenamientos desde cero en cada nuevo experimento.

El término universal no asume que un único modelo pueda resolver todos los casos sin adaptación, sino que se estudia hasta qué punto un detector previamente entrenado puede transferir conocimiento y cuánta adaptación necesita para funcionar en un dominio nuevo. Por ello, un detector útil debe combinar generalidad inicial, capacidad de adaptación y estabilidad ante cambios razonables del protocolo experimental.

Como modelo de referencia se ha usado ProPicker, un detector basado en segmentación 3D que permite dos modos de operación principales. El primero es la inferencia directa basada en el *prompt*, donde el modelo utiliza un ejemplo para condicionar la búsqueda de partículas similares. El segundo es la adaptación supervisada con un número limitado de datos, donde los pesos del modelo se ajustan al dominio objetivo. Esta estructura permite analizar la universalidad gradualmente, desde la transferencia inmediata hasta la especialización del modelo con poca supervisión.

Flujo experimental.

El flujo experimental se ha diseñado bajo criterios de reproducibilidad, integrando la gestión de datos, el entrenamiento y el análisis de resultados. Esta organización garantiza que factores como la conversión de coordenadas, el ajuste de contraste o la división de los conjuntos de datos no introduzcan sesgos en la comparación de resultados.

Por ello, el método se construye alrededor de controles explícitos que incluyen una división fija entre entrenamiento y validación, el uso de tamaños de *prompt* de 37^3 vóxeles establecidos por la herramienta y una evaluación con reglas geométricas consistentes en todos los experimentos.

Datos utilizados.

El estudio se apoya en dos fuentes de datos diferenciadas. Por un lado, EMPIAR-10988 aporta datos reales de ribosomas citoplasmáticos para validar la infraestructura técnica en un entorno biológico representativo. Este conjunto se utiliza como prueba inicial porque permite comprobar que el flujo completo de preparación, inferencia, adaptación del modelo y evaluación puede ejecutarse sobre tomogramas reales.

Por otro lado, el *dataset* sintético centrado en la tiroglobulina permite trabajar con un *ground truth* completo para analizar la sensibilidad a los *prompts* y la influencia de la simetría de la partícula bajo los efectos de la anisotropía. Este *dataset* contiene 25 tomogramas con aproximadamente 3327 partículas de la clase objetivo. Desde el segundo experimento se fija una división de 20 tomogramas para entrenamiento y 5 para validación.

La combinación de ambos tipos de datos responde a dos necesidades diferentes. Los datos reales permiten comprobar si el detector opera en un escenario cercano al uso práctico. Los datos sintéticos permiten aislar variables, disponer de anotaciones completas y estudiar con más detalle el comportamiento del sistema.

Metodología.

La metodología comienza con la preparación de los tomogramas y la generación de vectores de características mediante TomoTwin, una herramienta de localización de macromoléculas basada en similitud entre *embeddings*. Los volúmenes se normalizan, se convierten a la convención requerida por el detector y se acompañan de coordenadas expresadas de manera consistente. A partir de las anotaciones se extraen subtomogramas que actúan como *prompts*. Cada subtomograma se codifica con TomoTwin para obtener un vector de características que condiciona la red de segmentación de ProPicker.

Una vez condicionado el modelo, la inferencia produce mapas volumétricos de localización, los cuales no son todavía la salida final del sistema. Para obtener coordenadas discretas se aplica un posprocesado que incluye umbralización, *clustering* y cálculo de centroides. Después, las coordenadas predichas se comparan con el *ground truth*. Las métricas principales son precisión, sensibilidad y F_1.

Un aspecto relevante es la implementación de la inferencia *multi-prompt*, por la cual se realiza una agregación de las representaciones de cada *prompt* para reforzar los rasgos comunes de la partícula y mitigar el ruido o los errores de centrado. La agregación se realiza en el espacio latente y no sobre los vóxeles, permitiendo combinar información de varios ejemplos sin tener que alinear explícitamente las partículas en el volumen.

En los últimos experimentos se introduce una selección de *prompts* basada en características físicas del subtomograma y una selección en el espacio de rotaciones. Para la selección física se consideran propiedades como la densidad, la presencia de contenido estructurado y la ausencia de desplazamientos de intensidad que indiquen artefactos. Para la selección de la orientación se busca cubrir diversas rotaciones de la partícula. Dado que la tiroglobulina presenta simetría C_2, lo que implica que la estructura permanece idéntica tras una rotación de 180° sobre su eje principal, la distancia entre orientaciones se interpreta en el espacio cociente $SO(3)/C_2$.

Resultados experimentales.

Los experimentos muestran que la inferencia basada solo en *prompts* puede identificar partículas muy bien definidas, aunque suele ofrecer un rendimiento conservador. En EMPIAR-10988, el modelo base de ProPicker obtiene una precisión de 0.7049 pero un *recall* de 0.1966, con un valor F_1 de 0.3074. Este resultado indica que el modelo reconoce parte de la señal proveniente de los ribosomas, pero deja sin detectar muchas partículas. Tras ajustar el umbral de detección, el equilibrio mejora ligeramente y el F_1 sube a 0.3492. Sin embargo, el cambio más relevante aparece con la adaptación del modelo, que eleva el valor F_1 a 0.5714, lo que demuestra que la adaptación es necesaria para alcanzar unas prestaciones aceptables. La conclusión del primer experimento es que la transferencia directa contiene información útil, pero el simple ajuste de la sensibilidad no compensa la falta de adaptación al dominio.

En el conjunto sintético el cambio de dominio es más severo. Hemos pasado a trabajar con tiroglobulinas, las cuales se ven más afectadas por el ruido y las distorsiones anisotrópicas debido a su pequeño tamaño y estructura. El modelo base prácticamente no consigue localizar la tiroglobulina y obtiene valores de F_1 cercanos a cero. Tras la adaptación supervisada, el rendimiento medio alcanza 0.891. Este salto supone una clara mejora con respecto a los resultados casi nulos del modelo base en ese dominio. También muestra que ProPicker puede adaptarse de forma eficaz cuando recibe datos coherentes del objetivo.

El análisis de eficiencia indica que el sistema alcanza buenos resultados con apenas cuatro tomogramas de entrenamiento y que la estrategia *multi-prompt* es eficaz para reducir los falsos positivos. El experimento incremental entrena modelos con 1, 2, 4, 8, 12, 16 y 20 tomogramas manteniendo la misma validación. La curva crece con rapidez y se estabiliza después de los primeros incrementos. El mejor resultado observado aparece con 12 tomogramas e inferencia *multi-prompt*, alcanzando $F_1 = 0.919$.

No obstante, el último experimento reveló que la elección del *prompt* sigue siendo un factor crítico. Se generó una población amplia de *prompts* a partir de tomogramas de entrenamiento y se evaluó su efecto con un mismo *checkpoint*. Los resultados muestran que muchos *prompts* mantienen un rendimiento alto, pero algunos provocan una activación insuficiente del detector. El fallo principal es una activación insuficiente sobre partículas reales, más que un exceso de falsos positivos. Los análisis no permiten explicar completamente estas caídas usando solo SNR, posición vertical, distancia al centro o hemisferio de orientación. Esto apunta a una interacción más compleja entre la calidad de la imagen, la anisotropía de la muestra, la orientación de la partícula y la representación latente generada por TomoTwin.

Interpretación conjunta.

La discusión de los resultados permite concluir que la inferencia directa puede proporcionar una señal de partida, pero la adaptación supervisada es una condición práctica necesaria cuando cambian los datos. Así, el modelo base funciona mejor como una inicialización con conocimiento previo que como una solución final. Esta diferencia evita una interpretación demasiado optimista de la universalidad. ProPicker no resuelve por sí solo todos los dominios, pero sí ofrece una base útil sobre la que entrenar con pocos datos.

La cantidad de anotación requerida es moderada en comparación con entrenar un modelo desde cero y la agregación de *prompts* mejora la robustez al controlar la aparición de falsos positivos. Los resultados muestran que no siempre es necesario llegar al máximo número de tomogramas disponibles para obtener un rendimiento competitivo. A partir de cierto punto los beneficios adicionales son menores y la selección de *prompts* o la estrategia de inferencia puede ser tan importante como añadir más datos. Por tanto, el *prompt* debe considerarse un elemento central del detector.

Recomendaciones de optimización.

El trabajo propone prioridades para un flujo de trabajo real, empezando por la consistencia en el tratamiento de los datos antes de comparar modelos. Los nombres de tomogramas, las rutas, las unidades de las coordenadas, el contraste y la división entre entrenamiento y validación deben quedar fijados desde el principio. Sin esta consistencia es difícil saber si una mejora procede del modelo o de un cambio accidental en el procesamiento.

Se recomienda evitar selecciones arbitrarias de ejemplos y preferir casos estructurados y diversos, además de aplicar una política de adaptación incremental. Una estrategia razonable consiste en comenzar con pocos tomogramas, evaluar pronto y añadir datos solo cuando la validación lo justifique. También es fundamental hacer visibles los parámetros de posprocesado, ya que los umbrales, las distancias de agrupamiento y las reglas de emparejamiento forman parte del detector efectivo.

Limitaciones.

Aunque los experimentos analizan transferencia, adaptación supervisada, eficiencia de datos y robustez de *prompts*, todavía no prueban una universalidad amplia entre partículas, tipos de muestra y protocolos de adquisición. Además, algunos umbrales proceden de la configuración de referencia de ProPicker y la robustez frente a la elección del *prompt* sigue sin resolverse por completo. El análisis de orientaciones se estudia en detalle para la tiroglobulina, pero no se extiende a otras simetrías moleculares. Por ello, las conclusiones muestran dónde ProPicker ya es útil y dónde aún requiere mejoras, y dejan abierta la validación en conjuntos reales más amplios y en escenarios multiclase.

Conclusión.

El trabajo concluye que ProPicker es una herramienta eficaz para la localización de partículas cuando se integra en un flujo que combine la adaptación supervisada limitada con un control de calidad de los ejemplos. Aunque el modelo base es informativo, no se puede considerar todavía un detector universal plenamente autónomo. Su rendimiento depende del dominio, de la calidad de los datos, de la adaptación y de la representatividad del *prompt*. Sin embargo, los resultados muestran que puede transformarse en un detector sólido con una cantidad moderada de anotación y con una metodología controlada.

El avance hacia sistemas más robustos requiere incorporar mayor diversidad de datos reales, selección automática de *prompts* y representaciones que consideren explícitamente las simetrías moleculares. En conjunto, el trabajo muestra que la universalidad implica reutilizar conocimiento previo, reducir la supervisión necesaria y mantener un comportamiento estable al cambiar de dominio. Esta lectura también orienta futuras comparaciones entre detectores bajo protocolos más realistas y reproducibles.

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Acronyms

AIKE Artificial Intelligence and Knowledge Engineering.

Cryo-EM Cryogenic electron microscopy.

Cryo-ET Cryogenic electron tomography.

DL Deep learning.

3D Three-dimensional.

CNN Convolutional neural network.

CTF Contrast transfer function.

EMPIAR Electron Microscopy Public Image Archive.

hTG Human thyroglobulin.

IoU Intersection over union.

MRC Electron microscopy volume file format commonly used for tomograms.

TP/FP/FN True positives, false positives and false negatives.

PCA Principal component analysis.

RNA Ribonucleic acid.

SART Simultaneous algebraic reconstruction technique.

SIRT Simultaneous iterative reconstruction technique.

SNR Signal-to-noise ratio.

STA Subtomogram averaging.

TM Template matching.

UMAP Uniform manifold approximation and projection.

WBP Weighted backprojection.

Chapter 1

Introduction

1.1. Motivation

Cryogenic electron tomography (cryo-ET) has become one of the most powerful techniques for observing macromolecular organization *in situ*. By reconstructing a three-dimensional (3D) tomogram from a tilt series of a vitrified specimen, cryo-ET preserves spatial context that is lost in purified single-particle pipelines. This advantage is especially relevant when the scientific question concerns intracellular organization, molecular crowding or the coexistence of several complexes within the same environment [1, 2].

The computational burden begins after reconstruction. Tomograms are noisy, anisotropic and large, and several downstream stages depend on knowing where the particles of interest are located. Subtomogram Averaging (STA) requires coordinate estimates that are accurate enough to seed the rest of the structural analysis workflow [3, 4]. This process suppresses noise and compensates for missing wedge artifacts by extracting, aligning and averaging thousands of 3D volumes (subtomograms) of the same macromolecule. Therefore, particle picking is one of the main gates that determine whether a cryo-ET study scales or stalls.

Manual annotation does not scale to current datasets, while classical methods such as template matching (TM) often require strong assumptions about the target structure and can become brittle when the particle changes conformation or local environment. These constraints motivate the search for more flexible detectors that can retain good performance under heterogeneous acquisition conditions and under low-label regimes.

This bachelor thesis is framed around that goal. It studies ProPicker [5], a state-of-the-art deep learning (DL) picker with *universal* purpose. ProPicker is a promptable segmentation model that can either work from a prompt alone or be adapted to a target particle with supervised data. The main value of this work lies in establishing a reproducible experimental workflow and in identifying where the current generation of promptable detectors is already useful and where it still breaks down.

1.2. Problem statement

Consider a cellular context represented as a 3D density map

$$D(\mathbf{x}) : V \subset \mathbb{R}^3 \to \mathbb{R}$$

where $\mathbf{x}$ denotes a spatial coordinate within the volume of interest V sampled by a tomogram [6]. We define a prompt $\mathcal{P}$ as a compact representation of the target particle, used as an example and obtained in this work from one or more centered subtomogram crops and their corresponding TomoTwin [7] embeddings.

Given D and $\mathcal{P}$, the goal is to estimate a finite set of particle detections

$$\mathcal{D} = \{(\mathbf{p}_i, s_i)\}_{i=1}^{N},$$

where $\mathbf{p}_i \in \mathbb{R}^3$ is the predicted particle centroid in voxel coordinates and $s_i \in \mathbb{R}$ is a confidence or response score derived from the predicted localization map. The detector should produce detections that match the unknown ground truth coordinate set

$$\mathcal{Y} = \{\mathbf{y}_j\}_{j=1}^{M},$$

under a spatial tolerance criterion, while maintaining high precision and recall despite cryo-ET noise, anisotropic resolution, dataset and acquisition differences, limited annotated tomograms and sensitivity to the chosen prompt representation.

In practice, the complexity of this problem stems from a combination of several adverse conditions:

- Very low SNR caused by dose limitations.
- Anisotropic resolution due to the missing wedge.
- Crowded cellular context.
- Limited annotations and strong cost of manual labeling.
- Computational constraints imposed by large 3D volumes.

This work focuses on whether a promptable detector remains useful when supervision is scarce, the particle target or dataset changes and prompt quality becomes part of the problem.

1.3. Research questions

The work is organized around the following research questions:

1. Can prompt-only inference with ProPicker provide useful transfer on real cryo-ET data before any additional supervised adaptation?
2. How severe is the performance drop when the detector is moved to a different particle target and dataset?
3. How much annotated data is required to recover strong performance after adaptation?
4. Does aggregating several prompts improve practical robustness with respect to a single prompt?
5. Once adaptation is effective, is the remaining variability driven mainly by tomogram difficulty or by prompt factors such as orientation and embedding quality?

1.4. Contributions

The main contributions of the work are:

1. A reproducible and easy to scale research repository that organizes preprocessing, prompt construction, model adaptation, inference and result export for cryo-ET particle picking using ProPicker; see Appendix B for the public repository link.

2. A real data proof of concept on EMPIAR-10988 showing that the inference with the base model is viable but insufficient for balanced performance.

3. A dataset-and-target transfer experiment using PolNet-synthesized [6] thyroglobulin data demonstrating that supervised adaptation is essential for maintaining model performance.

4. A data efficiency study that identifies a strong operating point with only 4 training tomograms and the best observed configuration at 12 tomograms with multi-prompt inference.

5. A prompt robustness analysis showing that residual instability is expressed mainly as prompt-dependent recall collapse and is only weakly explained by source SNR or simple rotation proxies taken in isolation.

1.5. Document structure

Chapter 2 reviews the cryo-ET localization problem and the background needed to understand promptable detectors. Chapter 3 describes the objectives, datasets, methodology, implementation choices and evaluation protocol. Chapter 4 reports the results of the four completed experiments and discusses their implications. Chapter 5 summarizes the conclusions and outlines the future work. The appendix documents the hardware and software environment used in the project.

Chapter 2

State of the Art

This chapter is organized from general context to the specific task studied in the bachelor thesis. It first reviews how cryo-ET data are acquired, then summarizes the main particle-picking formulations and finally focuses on promptable detectors, dataset choices and the evaluation issues that matter for the experimental chapters.

2.1. Acquisition and reconstruction workflow

Cryo-ET reconstructs a 3D volume from a series of projection images acquired at different tilt angles. Because the angular range is limited in practice, the Fourier space is sampled incompletely and the reconstruction exhibits the missing wedge artifact [1, 2]. This causes direction-dependent distortions and makes pattern recognition intrinsically harder than in isotropic 3D imaging settings.

At a workflow level, the acquisition and reconstruction process can be summarized in four steps. First, a tilt series of two-dimensional (2D) projections is acquired while the specimen is mechanically tilted. Second, the images are motion-corrected and aligned, usually with fiducials or alignment models that compensate for grid deformation. Third, a 3D tomogram is reconstructed, typically through weighted or filtered backprojection (WBP) or through iterative methods such as simultaneous iterative reconstruction technique (SIRT) and simultaneous algebraic reconstruction technique (SART). Fourth, the reconstructed volume is used for downstream tasks such as particle picking, segmentation or STA [8, 9]. Modern preprocessing packages such as Warp illustrate how motion correction, contrast transfer function (CTF) handling and reconstruction-adjacent preparation have become central parts of cryogenic electron microscopy (cryo-EM) and cryo-ET workflows [10]. This sequence is important because the localization problem studied in this bachelor thesis is not applied to raw particle images but to reconstructed volumes that already contain the acquisition footprint of the microscope and the reconstruction algorithm.

Figure 2.1 provides an overview of the full cryo-ET pipeline discussed in this chapter, from tilt-series acquisition to tomogram reconstruction, denoised slices and the different levels of difficulty encountered during particle picking.

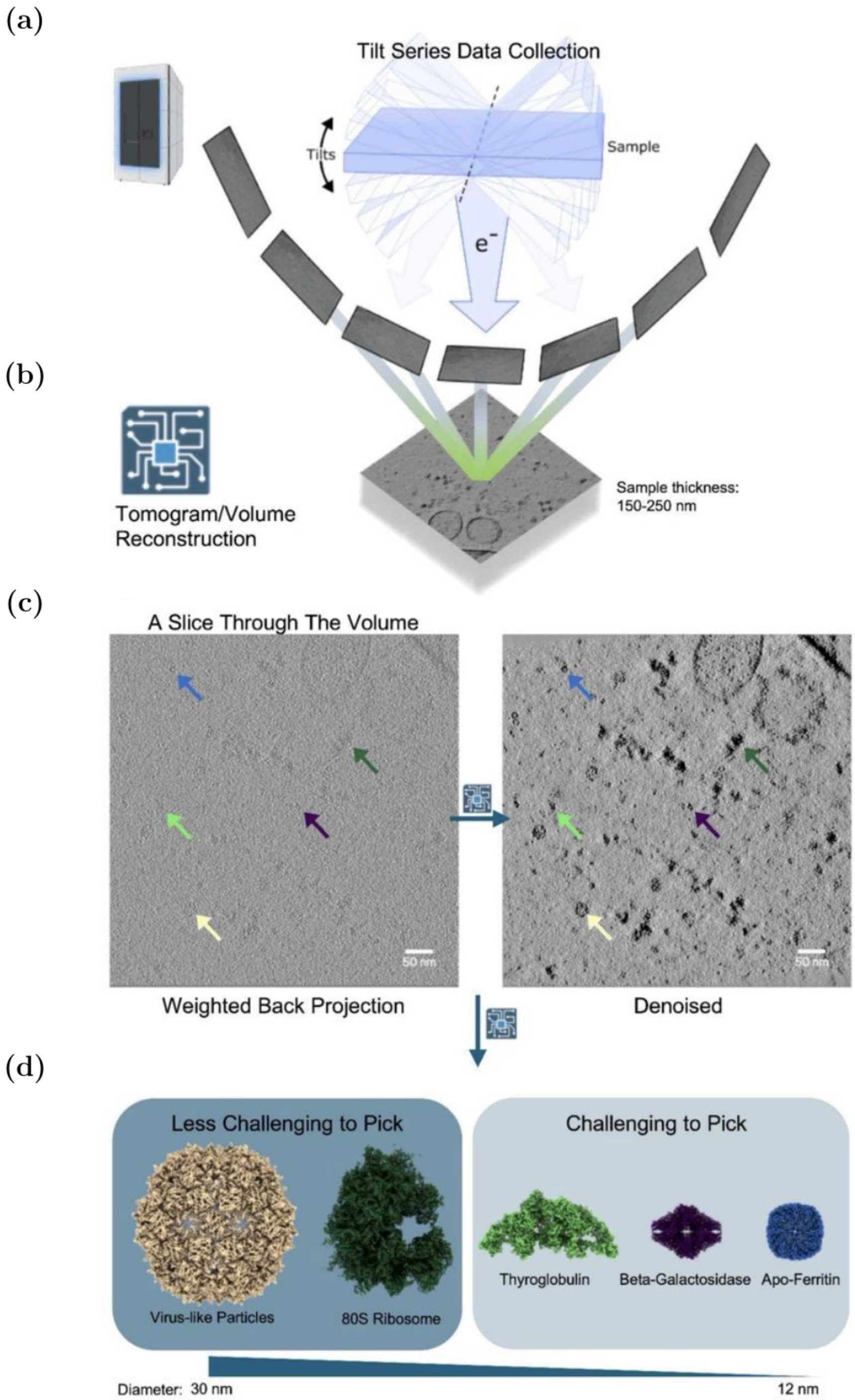


Figure 2.1: Overview of the cryo-ET workflow. (a) Tilt-series acquisition as the specimen is imaged over multiple angles. **(b)** Tomogram reconstruction from those projections. **(c)** Comparison between a weighted-back-projection slice and a denoised slice, where arrows indicate visible particles. **(d)** Particle classes with different picking difficulty, from larger virus-like particles and ribosomes to smaller targets such as thyroglobulin, β-galactosidase and apoferritin. Adapted from Figure 1 of Peck et al. [11].

2.2. Macromolecule localization and subtomogram analysis

Particle picking seeks to estimate the coordinates of particle centers inside the tomogram. Those coordinates are then used to extract subtomograms, which can be aligned, classified and averaged in order to recover a higher-SNR structural isotropic signal [3, 4]. A weak picker damages the entire downstream workflow: if recall is low, particles are missed; if precision is low, later stages waste time on FP or degrade the output due to contamination.

2.2.1. Classical localization strategies

Traditional cryo-ET localization methods fall roughly into two groups. TM, used for example in PyTom and Dynamo, compares the tomogram with a reference template across positions and orientations [12, 3]. Feature-based methods, including eigenvector segmentation and geometric boundary-feature detection, rely on filters, local maxima and hand-designed rules [13, 14]. TM remains the most widely accepted method, but it can be computationally expensive and it assumes that a sufficiently representative template is available. Hand-crafted strategies are usually more brittle when the target varies in appearance or when background clutter changes substantially.

2.2.2. Deep Learning formulations for 3D particle picking

Modern learning-based approaches usually reframe particle picking as one of the following problems:

- Voxel-wise segmentation or center heatmap prediction, as in DeepFinder and related CNN-based cryo-ET pickers [16, 27].
- 3D object detection on subvolumes, following the broader object-detection idea of localizing instances within candidate regions or dense prediction windows [18, 19, 20].
- Similarity-driven retrieval and matching in embedding space, as in TomoTwin structural data mining.

Architectures inspired by U-Net remain common because they combine local detail with contextual information and can be adapted to volumetric inputs [15]. DeepFinder is a representative example for macromolecular detection of this strategy: it uses a 3D convolutional segmentation network to produce class-wise score maps, then converts the segmentation output into particle coordinates through clustering [16]. More general object detection literature distinguishes methods that start from predefined reference boxes, known as anchor-based detectors, from anchor-free methods that predict object centers or keypoints more directly [17, 18, 19, 20]. In cryo-ET, however, the representation of interest is often a particle center rather than a tight 3D bounding box, so segmentation maps and center-localization maps fit better.

Figure 2.2 shows the CNN architecture used by DeepFinder together with its downstream localization workflow. The network follows an encoder-decoder design where convolutional blocks extract volumetric features, max-pooling increases the receptive field, up-sampling restores spatial resolution and skip concatenations transfer fine-scale information to the decoder. The trained CNN converts an input tomogram into a multiclass segmentation map, after which clustering turns connected segmented regions into particle coordinates.

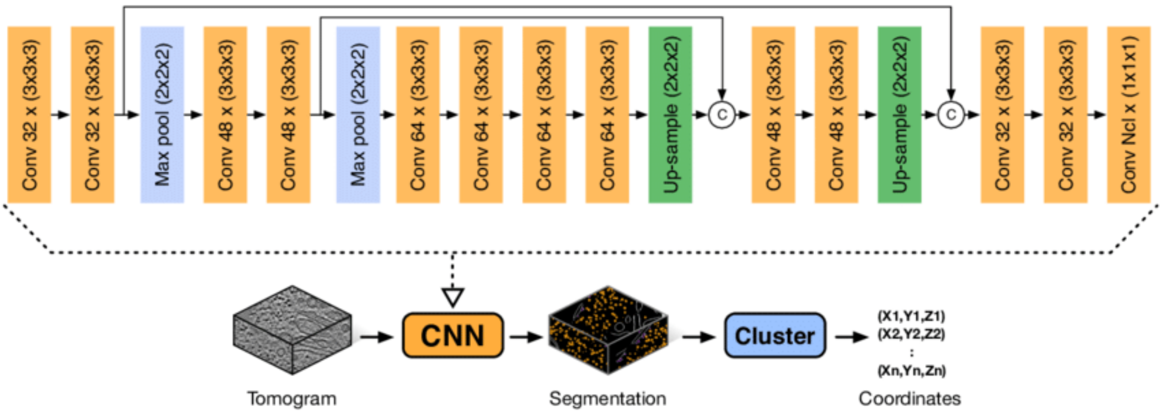


Figure 2.2: Architecture of DeepFinder. Orange blocks denote 3D convolutional layers, blue blocks denote max-pooling operations, green blocks denote up-sampling operations and circled C nodes denote concatenation of encoder and decoder feature maps. The final N_{cl}-channel convolution produces a multiclass segmentation map; clustering then converts segmented regions into particle coordinates. Adapted from Extended Data Figure 1c of Moebel et al. [16].

Despite the progress of generic 2D detectors [21], cryo-ET retains a clear volumetric character. Working with 3D patches is therefore advantageous, but it also imposes memory constraints that force careful tiling, padding and response aggregation.

2.2.3. Promptable detectors and the idea of universality

The notion of a universal detector in cryo-ET is tied to flexibility. A model that performs well on one fixed class but requires retraining from scratch for every new particle is useful, but not universal in any practical sense. Promptable detectors attempt to bridge that gap by conditioning a general model on a reference example of the target particle.

ProPicker is a recent example of this family. It combines a prompt encoder with a conditioned 3D segmentation network, producing a localization map from which coordinates are extracted [5]. This enables prompt-only inference for new particles and, when that is insufficient, supervised adaptation using data from the new target.

Figure 2.3 summarizes the workflow. A prompt crop is encoded into a particle-specific representation, the tomogram is processed by a conditioned segmentation model and the final particle coordinates are recovered from the predicted segmentation mask through post-processing.

This design follows broader promptable-vision trends, but cryo-ET makes the prompt itself a source of variability. Its orientation, local context, centering quality and embedding statistics can all affect the segmentation response. Universality therefore also depends on the stability of the conditioning mechanism.

2.2.4. From tilt-series acquisition to rotation-sensitive prompts

In an ideal isotropic 3D imaging system, rotating a particle produces a rotated copy of the same signal. Cryo-ET breaks that symmetry because the specimen is scanned over a limited angular range and around a single mechanical tilt axis. The resulting missing wedge removes direction-dependent Fourier information and creates anisotropic blur and elongation in the reconstructed tomogram [1, 2]. Consequently, two subtomograms

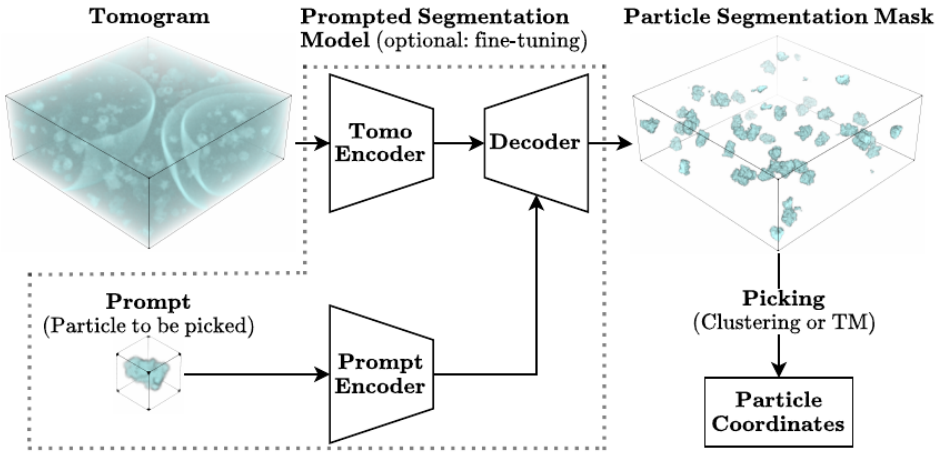


Figure 2.3: Overview of the ProPicker workflow. A prompt encoder extracts features from a reference particle, those features condition the tomogram segmentation model and the predicted segmentation mask is converted into particle coordinates by a downstream picking step. Adapted from Figure 1 of Wiedemann et al. [5].

containing the same macromolecule at different orientations are not related by a clean rigid rotation alone. This can be written schematically by defining a particle crop as a local scalar field

$$P(\mathbf{x}) = D(\mathbf{x}_0 + \mathbf{x}), \qquad \mathbf{x} \in \Omega \subset \mathbb{R}^3,$$

where Ω is the crop support around the particle center $\mathbf{x}_0$. A rigidly rotated version of the same particle is then

$$G(\mathbf{x}) = P(R^{-1}\mathbf{x}),$$

and its limited-angle reconstructed appearance can be approximated as

$$Y(\mathbf{x}) \approx \mathcal{A}_{\text{mw}}(G)(\mathbf{x}) + \eta(\mathbf{x}).$$

Here R is the particle orientation, $\eta(\mathbf{x})$ is noise and $\mathcal{A}_{\text{mw}}$ is the acquisition-reconstruction operator induced by limited-angle tomography. The key point is that $\mathcal{A}_{\text{mw}}$ is not rotationally neutral: rotating the object before reconstruction is not equivalent to rigidly rotating the reconstructed volume. This anisotropy is precisely why missing-wedge compensation methods such as IsoNet are needed [22].

This has a direct impact on promptable detection. A prompt crop includes the target particle together with missing-wedge effects, local reconstruction context, centering errors and residual contrast distortions. A geometrically favorable prompt can therefore yield a more stable representation than another prompt from the same class at a less favorable orientation.

The model architecture adds a second difficulty. Standard CNNs are naturally equivariant to translations, but not to arbitrary rotations [23]; in 3D, rotation equivariance requires specialized constructions such as steerable convolutions [24]. ProPicker gains flexibility by conditioning the detector on a prompt, but that does not make the overall pipeline rotation-invariant [5].

The specific target used in the synthetic dataset adds another source of rotational ambiguity. Human thyroglobulin is a homodimer with global C2 symmetry, and the reported cryo-EM structure is approximately $250 \times 160 \times 110$ Å, with the shortest dimension aligned with the C2 axis [25]. Two poses that differ by a 180° rotation around that axis are physically equivalent, so a pose-sensitive system naturally operates over $SO(3)/C_2$ rather than pure $SO(3)$.

For thyroglobulin, the problematic rotations are therefore of two kinds: symmetry-equivalent rotations, which should ideally be collapsed by the loss or evaluation metric, and acquisition-amplified ambiguities, where discriminative pose differences are degraded by the missing wedge. This is why physically plausible 3D augmentations, symmetry-aware losses and matched missing-wedge training are attractive strategies in cryo-ET [22, 24, 9].

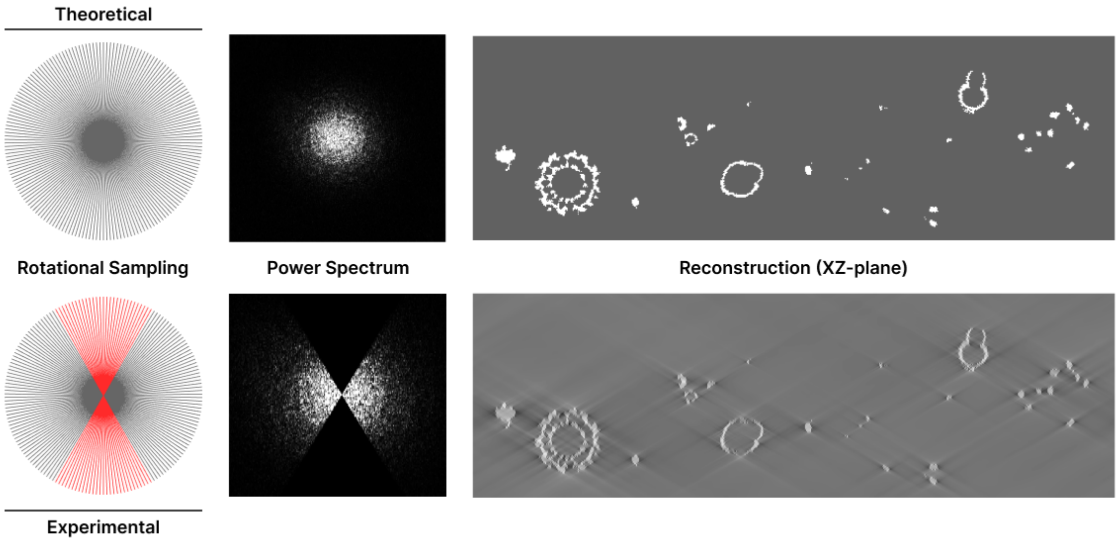


Figure 2.4: Wedge-like reconstruction problem in cryo-ET. The **theoretical** case shows complete rotational sampling and a reconstruction without missing angular information, while the experimental case shows limited sampling, a missing wedge in Fourier space and elongation artifacts in the reconstructed XZ plane. In the **experimental** rotational-sampling panel, the red angular sectors indicate directions that are not acquired because the specimen cannot be tilted through a full 180° range; these unsampled directions correspond to the dark wedge-shaped gaps in the power spectrum. This acquisition-level anisotropy explains why rotated subtomograms are not simple rigid copies of one another and why prompt orientation can affect downstream particle picking. Adapted from the CryoET Data Portal documentation [26].

2.3. Real and synthetic datasets in cryo-ET

Real and synthetic tomograms serve complementary purposes. Real datasets capture the complexity of laboratory conditions, reconstruction artifacts and biological variation. Synthetic datasets, by contrast, provide full ground truth and allow controlled experiments on noise, density, class distribution and missing-wedge-related distortions. A serious evaluation program benefits from both. That is precisely why this work combines EMPIAR-10988 and PolNet synthetic data. The real dataset answers whether the detector is useful in a realistic scenario. The synthetic dataset answers more controlled questions about adaptation, data efficiency and prompt robustness, because it allows stable train/validation splits and exhaustive metric computation.

Chapter 3

Objectives and Methodology

This chapter joins the objectives with the methodological decisions that make the later results comparable. The goal is to describe not only what was evaluated, but also how the code developed enforces stable data splits, coordinate conventions, prompt construction, training schedules, inference outputs and evaluation rules.

3.1. Objectives

The bachelor thesis investigates the practical value of ProPicker as a foundation for universal macromolecule localization in cryo-ET. Rather than focusing on optimizing leaderboard metrics, the project objective is to study the current limits of ProPicker and translate these findings into an optimized pipeline. This approach aims to maximize the model's performance through systematic experimentation without altering the architecture.

The specific objectives are:

1. Build a reproducible repository that can support repeated adaptation and inference studies across several datasets.
2. Validate the complete workflow on real cryo-ET data.
3. Quantify transfer effects on a synthetic dataset with full ground truth.
4. Measure data efficiency and compare prompt aggregation strategies.
5. Study the robustness limits of the current prompt-conditioning mechanism.
6. Establish a project foundation that scales easily to new experiments in future work.

3.2. Datasets

3.2.1. Experimental data: EMPIAR-10988

The real dataset uses the EMPIAR-10988 ribosome case already supported by the ProPicker examples [27]. Ribosomes are abundant, large and well-characterized cellular macromolecular machines responsible for protein synthesis, which makes them a natural first target for pipeline validation. In the repository, the experiment focuses on the **cytosolic ribosome** target. Training is performed on tomogram `TS_029` and validation on `TS_030`. The workflow uses a centered crop with a `crop_delta` of 64 (half-width in voxels), which yields a controlled subvolume of 128^3 voxels around the region of interest. This setup is a proof of concept on real data, not a full-scale evaluation over the entire EMPIAR-10988 collection. The primary goal was to verify the pipeline, from prompt extraction to adapted-model inference, under realistic data conditions while keeping the experiment small enough for careful debugging. Furthermore, ribosomes were selected as the initial target because they are relatively easy to pick. Their large size, distinctive symmetrical form and high contrast relative to the background make them ideal for validating new localization pipelines.

Because Experiment 1 (EXP1) focuses on cytoplasmic ribosomes, Figure 3.1(a) is included as a structural reference for the target macromolecule. It is illustrative rather than dataset-derived, but it helps anchor the discussion around the particle class being localized in the tomograms.

3.2.2. Synthetic dataset: PolNet

The main synthetic dataset consists of PolNet synthetic tomograms. The completed experiments focus on the **thyroglobulin** class across 25 tomograms and approximately 3327 labeled thyroglobulin particles in total. From Experiment 2 (EXP2) onward, the dataset follows a fixed `20/5 split`, using 20 tomograms for training and 5 tomograms for validation. Keeping the validation set fixed is essential because EXP2, Experiment 3 (EXP3) and EXP4 are intended to be directly comparable.

Figure 3.1(b) shows the global form of thyroglobulin. The particle is elongated, non-spherical and smaller than the ribosome, so its weaker electron-density signal and orientation-dependent reconstructed appearance make thyroglobulin a harder target. Under limited-angle cryo-ET acquisition, the missing wedge can therefore remove different parts of the structural signal depending on how the particle is rotated in the tomogram. This makes the target a useful case for studying form ambiguity, prompt orientation effects and rotation sensitive failures.

3.3. Method overview

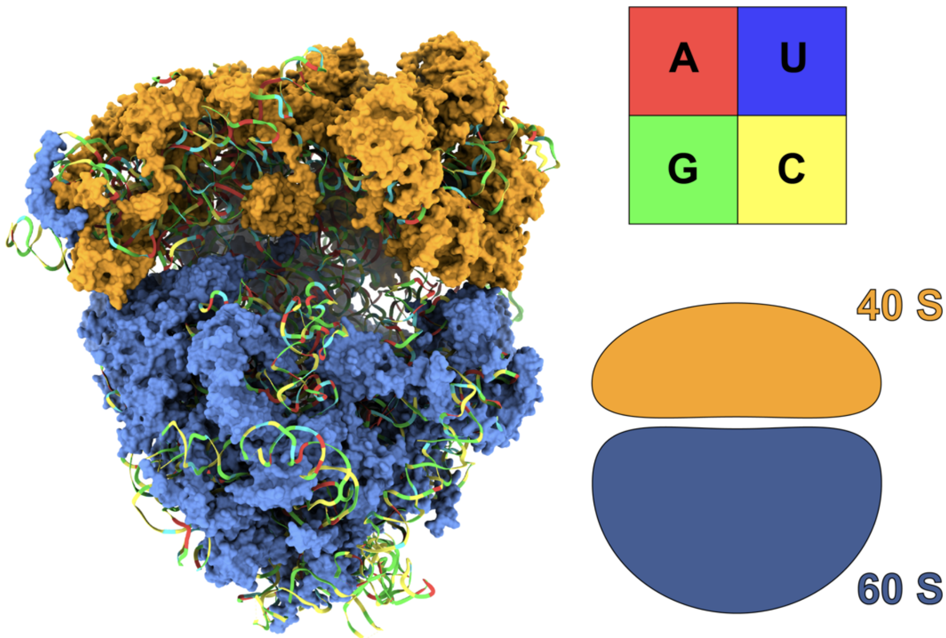
3.3.1. Adopted detector and design rationale

The detector used throughout the work is ProPicker. The choice is motivated by its promptable design and by its ability to support two complementary modes:

- **Prompt-only inference**, which tests how much transfer to a new particle target and dataset can be obtained without additional supervised adaptation. It provides a direct way to evaluate whether the base model already contains useful general prior information that can be reused for new particle classes with minimal intervention.
- **Particle-specific adaptation**, which tests how efficiently the same model can adapt when transfer alone is insufficient. It allows the study to measure how much supervision is needed to recover strong performance once the detector faces a substantial change in particle type, dataset or acquisition conditions.

This makes ProPicker a suitable tool for studying universality, as it allows us to systematically evaluate how far a single detector can generalize across particle types through prompting alone, and how efficiently it can adapt when supervised target-specific training data are required.

(a)



(b)

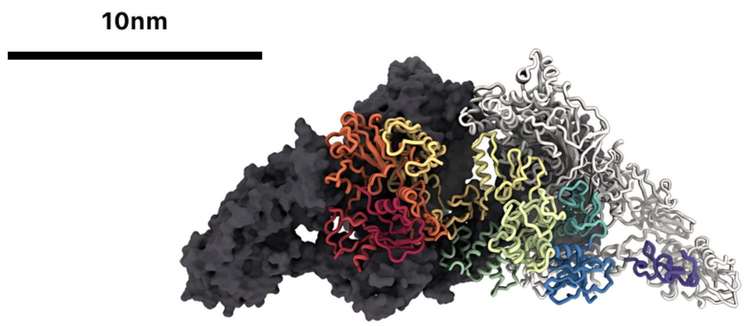


Figure 3.1: Structural references for the two particle targets used in the investigation. (a) Illustrative 80S human ribosome model used as a visual reference for the EMPIAR-10988 cytoplasmic-ribosome experiment. The information on the right summarizes the ribosome composition: the colored A, U, G and C boxes identify the four RNA bases, and the 40S and 60S schematics indicate the small and large eukaryotic ribosomal subunits. **(b)** Type 1 repeats in the human thyroglobulin structure. One human thyroglobulin (hTG) monomer is shown as a charcoal surface, while the other monomer highlights the individual type 1 repeats as colored ribbons. Approximate scale bars: 10 nm. Reproduced from Ykust [28], licensed under Creative Commons Attribution (CC BY) 4.0. Adapted from Figure 2 of Adaixo et al. [25].

3.3.2. Prompt construction and inference

Each prompt is built from a subtomogram centered on a labeled particle. The prompt encoder transforms the crop into an embedding that conditions the segmentation network. Inference then produces a localization map over the target tomogram, after which clustering and thresholding convert the response into discrete coordinates.

The work studies three prompt regimes:

1. One fixed prompt per experiment, used as the simplest baseline.

2. A multi-prompt embedding obtained from several instances to stabilize inference.

3. Rotation-diverse prompt sets, used to probe orientation sensitivity.

Prompt embedding with TomoTwin. In this context, an embedding is a compact numerical vector that represents the visual and structural content of a prompt subtomogram. TomoTwin produces this vector by passing a crop of 37^3 voxels through a pretrained 3D encoder and taking the encoder output as the prompt descriptor. The result is no longer a voxel volume, but a latent representation that ProPicker can use to condition its segmentation network toward the particle class shown by the prompt.

The implementation became progressively more selective as the investigation advanced. In EXP1 and EXP2 the prompt is a fixed, manually validated positive instance, because the objective there is to measure transfer and supervised adaptation, not prompt search. In EXP3 the study introduces a robust prompt-ranking function in order to avoid a naive strategy based only on local variance. The physical-quality score used in that selector is

$$q_i = w_f \log(f_i) + w_c \log(c_i) - w_{dc} d_i,$$

where f_i is a mid-frequency to high-frequency Fourier power ratio, c_i is a center-to-border intensity ratio and d_i is a direct-current (DC) penalty term that penalizes offset-dominated crops. The default weights are $w_f = 1.0$, $w_c = 0.9$ and $w_{dc} = 0.6$. The selector first ranks physically plausible candidates, keeps the best $k = 8$, and can in principle be extended with an embedding-consensus term. In the actual EXP3 single-prompt workflow, the final prompt was selected from the physical score alone.

The multi-prompt branch of EXP3 follows a different logic. It extracts ten candidate subtomograms, computes their individual embeddings with TomoTwin and represents the particle class in two ways: through the first embedding as a single-instance prompt and through the arithmetic mean of all ten embeddings as a multi-instance prompt.

Formation of the multi-prompt embedding

The multi-prompt is formed before inference, from the already extracted prompt subtomograms of 37^3 voxels. Let $\{s_i\}_{i=1}^{10}$ be the available candidate crops for the target particle class. Each s_i is still a real 3D crop centered on one annotated thyroglobulin particle. The candidates are first ranked by local intensity variance, but the final set is not chosen as the top ten by that score. Instead, ten indices are sampled uniformly across the variance-ranked list. This keeps prompts with different contrast and structural strength while reducing the risk that the representation is dominated by a few noisy or unusually high-variance crops.

Each selected crop s_i is then passed independently through the TomoTwin prompt encoder,

$$e_i = \mathrm{TomoTwin}(s_i),$$

which produces one embedding vector per selected particle instance. At this point the information is no longer represented as a visible subtomogram. The crop has been mapped into a latent feature space where nearby vectors should correspond to particles with similar structural appearance. The «mixture» of the ten thyroglobulin examples is therefore not performed in voxel space and does not create a new averaged 3D particle volume.

The operational multi-prompt representation is instead the arithmetic mean of the ten instance-level embeddings,

$$e_{\text{multi}} = \frac{1}{10}\sum_{i=1}^{10} e_i.$$

This averaged vector is saved in the same JavaScript Object Notation (JSON) format expected by ProPicker, under the thyroglobulin class key. Therefore the detector still receives a single conditioning input at inference time, but that input is a consensus feature representation of several real particles rather than one isolated crop. It can be interpreted operationally as a more representative prompt for the particle class: common thyroglobulin features are reinforced, while instance-specific effects such as noise, centering error, local background and unusual orientation are partially smoothed out. The single-instance prompt used for comparison is saved from the first selected embedding, while the full set of individual embeddings is kept separately for analysis.

EXP4 adds a third layer of control by selecting prompts in quaternion space. After quality filtering, prompts are chosen greedily to maximize the minimum angular separation

$$\theta(q_i, q_j) = 2 \arccos\left(|q_i \cdot q_j|\right),$$

with a target minimum distance of 30°. This is the operational definition of rotational diversity used throughout the robustness study.

3.3.3. Preprocessing and annotation strategy

The preprocessing workflow is deliberately conservative because the work aims to compare experimental factors rather than to re-optimize the entire toolchain at every exploratory stage. The shared pipeline can be summarized as follows:

1. Input tomograms are read in MRC format and standardized into a shared data layout, with consistent file naming and coordinate conventions across experiments.
2. Contrast is inverted when needed so that the synthetic and real crops follow the ProPicker/TomoTwin convention of bright particles on dark background.
3. Particle coordinates are expressed in the coordinate convention required by the detector and the evaluator.
4. Prompt subtomograms of 37^3 voxels are extracted around labeled particles.
5. Prompt embeddings are generated with the TomoTwin encoder bundled with the repository workflow.
6. The preprocessing stage produces normalized tomograms, labels and train/test definitions that are reused consistently in adaptation and inference.

For experiments that use the PolNet data, the pipeline contains two additional transformations that are scientifically important. First, the annotation metadata are reconciled with the local identity of each tomogram so that the dataset remains internally consistent. Second, the thyroglobulin coordinates are converted from Å to voxel units using the voxel-size metadata read from the tomograms. This solves the common error of comparing voxel-based predictions with physical coordinate annotations, a major hurdle for pipeline reproducibility.

3.3.4. Model adaptation pipeline

Model adaptation follows the standardized ProPicker training protocol used throughout the study. Across the PolNet experiments, the main shared settings are:

- Patch size: 72 voxels.
- Padding size: 12 voxels.
- Label diameter: 21 voxels.
- Learning rate: 10^{-3}.
- Batch size: 2 for the synthetic experiments.

EXP1 differs slightly because it follows the ribosome proof-of-concept regime, including a batch size of 8 and 75 epochs for the centered 128^3 voxels crop. In the synthetic study, the incremental schedule of EXP3 adapts the number of epochs to the number of training tomograms in order to avoid unfairly undertraining the smallest subsets. The implemented schedule is $100, 80, 50, 35, 25, 22$, and 20 epochs for training sets of $1, 2, 4, 8, 12, 16$, and 20 tomograms respectively.

An important methodological detail is that all long runs are defined from the same preprocessing products, the same split definitions and the same evaluation rules before training starts. It removes path-dependent and setup-dependent variation before training begins and ensures that differences between experiments can be interpreted as scientific differences rather than procedural ones.

3.3.5. Monitoring, exports and visualization

The framework supports three complementary inspection levels: scalar metrics for comparison, full localization maps for qualitative review and prompt metadata tables for later robustness analysis. Localization maps can be represented both as tensor outputs for numerical post-processing and as MRC volumes for volumetric inspection in ParaView. This separation between scalar, volumetric and prompt-level evidence is important because no single inspection level is sufficient on its own to explain detector behavior in cryo-ET.

For this reason, training and inference runs are not treated as disposable executions. Checkpoints, prompt embeddings, predicted coordinates, localization maps and summary tables are stored by experiment and configuration. This is especially important in EXP3 and EXP4, where several training increments, prompt variants and candidate model states have to remain auditable under the same validation protocol.

3.4. Experimental protocol

3.4.1. Scientific workflow and reproducibility controls

The scientific workflow used in the work follows the same sequence in all four experiments:

1. Formulate a narrowly defined experimental question and an operational hypothesis.
2. Freeze the relevant split, preprocessing rules and metric implementation before launching long runs.
3. Isolate the factor under study so that only one main axis changes at a time.
4. Execute training or inference under standardized conditions that save checkpoints, configurations and outputs in deterministic result sets.
5. Perform quantitative evaluation first and qualitative or failure analysis second.

This order is central to the credibility of the results. EXP1 changes model adaptation and inference threshold, not the dataset. EXP2 changes the particle target and dataset while keeping the new PolNet split fixed. EXP3 changes training-set size and prompt aggregation while preserving the split and the fixed procedure used to match predictions to ground truth and compute metrics. EXP4 changes prompt orientation and checkpoint family while reusing the same validation tomograms. The framework therefore acts not only as storage, but as an experimental control mechanism.

3.4.2. Baselines and controlled comparisons

The experimental program is cumulative. Table 3.1 summarizes the completed experiments and the main comparison made in each one.

Exp.	Dataset	Goal	Main comparison
EXP1	EMPIAR-10988 ribosomes	Validate the full workflow on real data	Base ProPicker vs threshold-tuned base vs fine-tuned model
EXP2	PolNet synthetic, thyroglobulin	Measure severe transfer failure and recovery through adaptation	Base ProPicker vs adapted model on a fixed 20/5 split
EXP3	PolNet synthetic, thyroglobulin	Measure data efficiency and prompt aggregation effects	Incremental training sizes and single-prompt vs multi-prompt inference
EXP4	PolNet synthetic, thyroglobulin	Study residual robustness after adaptation	Base vs adapted checkpoint families and prompt-level rotational variability

Table 3.1: Overview of the completed experiments.

3.4.3. Metrics

The main scalar metrics are

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN}, \qquad F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

where TP, FP and FN denote correct detections, spurious detections and missed particles, respectively. Precision measures the fraction of reported detections that are correct and is therefore useful for quantifying contamination of the picked subtomogram set, whereas recall measures the fraction of real particles that are recovered and is therefore useful for quantifying how much potentially valuable structural signal is being lost. The F_1 score is the harmonic mean of precision and recall, so it increases only when the detector maintains a reasonably balanced operating point instead of improving one term while collapsing the other.

In cryo-ET particle picking, F_1 is commonly used as the main summary metric because the task is strongly imbalanced and because both kinds of error are costly for downstream analysis: too many FP waste inspection effort and contaminate STA, while too many FN reduce the amount of recoverable structural information [3, 4]. For that reason, a global accuracy measure would be misleading, whereas F_1 provides a compact description of practical detector quality while precision and recall remain available to indicate the dominant failure mode.

These metrics are inseparable from post-processing, because clustering, thresholding, coordinate extraction, geometric matching and size or overlap filters all influence the final counts of TP, FP and FN.

3.4.4. Hyperparameter policy

The objective of this work is not to perform a broad hyperparameter search. Once a stable setup was found, the policy was to keep it fixed whenever possible and only change one axis of variation at a time. This decision is especially important for EXP3 and EXP4, where the point is to compare training set size and prompt properties, not to co-optimize all training settings jointly.

Parameters like patch size, padding, prompt size, contrast inversion and the PolNet validation split are kept constant once they become part of the shared synthetic dataset. Only the factors required by the research question like threshold calibration in EXP1, supervised adaptation in EXP2, sample efficiency and prompt aggregation in EXP3 and rotational prompt diversity in EXP4 are allowed to move.

Chapter 4

Experimental Results

This chapter documents the four experiments conducted to answer the hypotheses posed in this document. The first experiment tests the model's performance with real data from EMPIAR-10988 ribosomes, the second measures transfer to synthetic PolNet thyroglobulin, the third studies the efficiency and learning capacity of ProPicker and the fourth investigates the model's weaknesses based on the results obtained. Each section corresponds to a single experiment and is named according to its main result.

4.1. Fine-tuning turns real-data transfer into balanced ribosome detection

We want to know whether ProPicker contains enough transferable prior information to detect ribosomes on real cryo-ET data before any adjustment. If the pretrained detector has learned useful information from the structures within cryo-ET tomograms, it should show detectable signal even before adaptation, but this signal is expected to be poorly calibrated and biased toward precision rather than recall. The question is whether lightweight fine-tuning can convert a conservative detector into a balanced one.

For this, the experiment uses the `cyto_ribosome` case of EMPIAR-10988, with `TS_029` for training and `TS_030` for validation under `crop_delta` of 64 voxels. The methodology is carried out in two stages. First, the base model is run in prompt-only mode and its localization map is transformed into coordinates through cluster post-processing. It calibrates:

- The binarization threshold, which controls the precision-recall balance through the size of the activated clusters.
- The minimum cluster size, initialized near 0.1 times the expected ribosome ball volume and the maximum cluster size, initialized near 1.2 times the same volume.

The evaluator then applies the same criterion used in the reference workflow, counting a detection as correct when it reaches IoU ≥ 0.6. Second, the model is fine-tuned on the cropped ribosome setting with the same prompt construction and the same evaluation rule. This preserves the scientific comparison: only the model adaptation changes, not the data split or the evaluator. In the repository, this protocol is implemented by the EMPIAR fine-tuning and inference scripts, which handle the real-data crop, contrast inversion, ProPicker training wrapper and inference wrapper.

Figure 4.1 shows the key finding of the real-data experiment. The pretrained model behaves conservatively. It finds ribosomes with decent precision, but its recall is too low for operational use. Threshold tuning improves F_1 slightly, which confirms that part of the issue is calibration. Still, the decisive improvement comes from fine-tuning: the F_1 score rises from the 0.31 to 0.35 range into 0.5714, with recall more than doubling relative to the base model.

Setting	Precision	Recall	F_1	TP	FP	FN
Base ProPicker	0.7049	0.1966	0.3074	547	229	2236
Base with optimized threshold	0.5125	0.2648	0.3492	737	701	2046
Fine-tuned model	0.5677	0.5752	**0.5714**	1044	795	771

Table 4.1: EXP1 quantitative comparison. Threshold tuning changes calibration and raises F_1 only modestly, whereas fine-tuning raises recall from 0.1966 to 0.5752 while preserving usable precision.

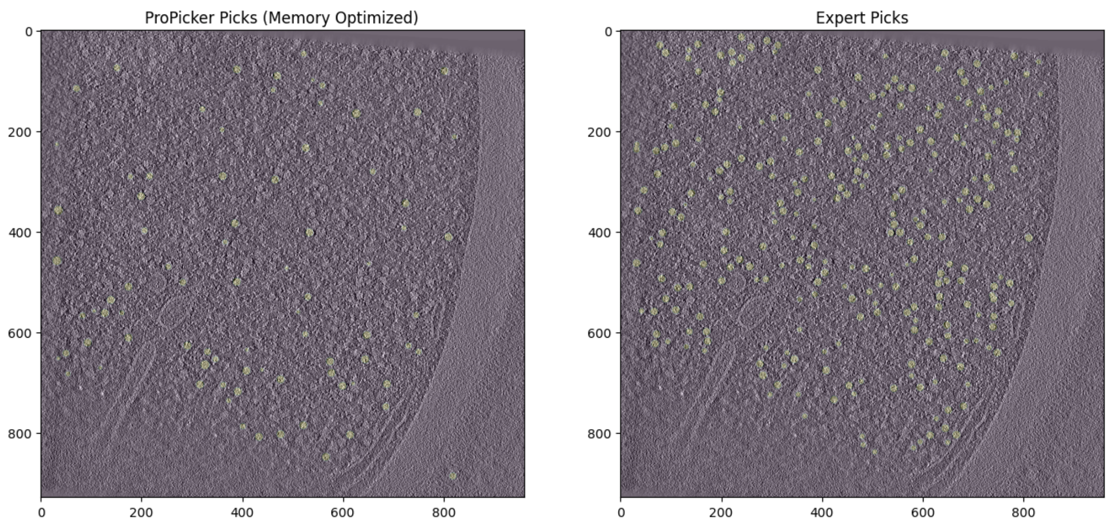


Figure 4.1: EXP1 fine-tuning effect on real EMPIAR-10988 ribosome data. Qualitative comparison between detector picks and expert annotations on the same tomographic slice. The detector localizes ribosome-like regions, but the denser annotation panel explains why recall is the limiting factor before adaptation.

The result is consistent with three effects that must be separated in the rest of the document. First, the promptable formulation carries useful information even on real data. Second, threshold calibration matters enough to deserve explicit treatment. Third, calibration alone does not close the gap: the real change of regime comes from fine-tuning, which lifts recall from 0.1966 to 0.5752 and reduces the false-negative count from 2236 to 771. Figure 4.1 and Table 4.1 show the same pattern in spatial and quantitative form: the detector finds plausible particles, but it undersamples the expert annotation field until the model is adapted.

4.2. Supervised adaptation is required for the synthetic thyroglobulin target

The next question is whether the signal observed in EXP1 survives a much stronger change in particle target and dataset. In this case, the experiment moves from real EMPIAR ribosomes to synthetic PolNet thyroglobulin. If promptability alone is not enough to bridge this change, the pretrained model should collapse, while supervised adaptation on a controlled target split should recover high performance if the rest of the pipeline is coherent.

For this, the experiment freezes the now-standard 20/5 split of the 25 PolNet tomograms and keeps that validation set unchanged in all later synthetic experiments. The preprocessing stage performs the following transformations in a reproducible order:

1. Reconcile annotation metadata with the local identity of each tomogram.
2. Isolate only the labels corresponding to `thyroglobulin`.
3. Convert particle coordinates from Å to voxels.
4. Extract prompts of 37^3 voxels.
5. Invert contrast to match the ProPicker/TomoTwin convention.
6. Generate the normalized inputs required by the detector.

The evaluation stage uses the same validation tomograms in all comparisons and matches predicted centers to ground truth with a threshold of half the prompt size. The corresponding PolNet scripts implement the 20/5 split, voxel-space coordinate conversion, contrast inversion and inference export used throughout the synthetic experiments.

Validation tomogram	Precision	Recall	F_1	TP	FP	FN
5 (SNR = 1.66)	0.894	1.000	**0.944**	127	15	0
6 (SNR = 1.17)	0.895	0.992	0.941	119	14	1
7 (SNR = 1.13)	0.868	0.985	0.923	131	20	2
8 (SNR = 0.57)	0.565	0.939	0.706	108	83	7
9 (SNR = 1.28)	0.888	1.000	0.941	127	16	0
Mean adapted-model performance	0.822	0.983	**0.891**	–	–	–
Mean base performance	0.009	0.008	0.008	–	–	–

Table 4.2: EXP2 quantitative comparison by validation tomogram. Four tomograms remain near $F_1 \approx 0.94$, while tomogram 8 falls to $F_1 = 0.706$. The base performance remains near zero, whereas the adapted model reaches mean $F_1 = 0.891$.

The contrast in Table 4.2 is the strongest result of the investigation. The pretrained model essentially fails on this new target and dataset, with mean precision, recall and F_1 all close to zero. In practice, the base detector does not transfer from its original training distribution to PolNet thyroglobulin, it produces too few useful matches and leaves almost the complete validation particle field undetected.

After supervised adaptation using the 20 training tomograms, the same pipeline changes regime. This is therefore an adaptation step for the synthetic thyroglobulin target rather than a minimal fine-tuning test. The adapted model reaches mean precision 0.822, mean recall 0.983 and mean $F_1 = 0.891$ on the fixed five-tomogram validation subset. The high recall is especially important because it shows that the model has learned to activate on the new synthetic particle class rather than only improving confidence calibration. The remaining errors are not distributed uniformly across the validation set. Four tomograms reach F_1 values around 0.92–0.94, while tomogram 8 drops to $F_1 = 0.706$ because its low SNR means weaker particle signal relative to noise, reducing precision to 0.565 despite recall remaining high at 0.939.

The representative spatial comparison in Figure 4.2 complements the table by showing how the adapted model recovers the particle distribution that the base model mostly misses.

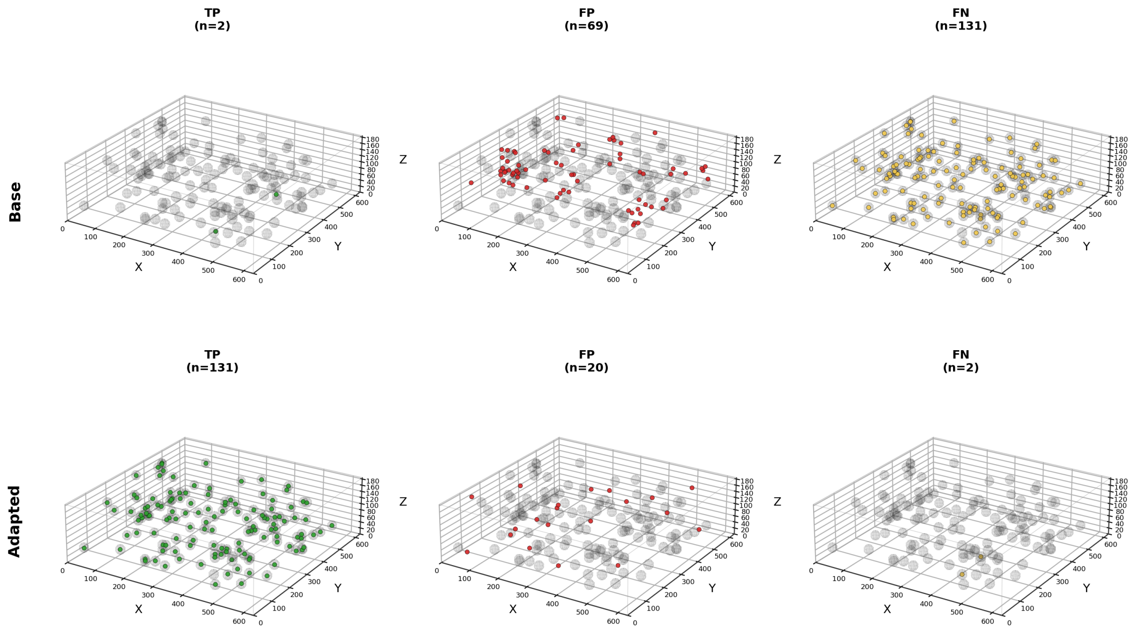


Figure 4.2: EXP2 spatial recovery on the fixed PolNet validation split. Three-dimensional TP, FP and FN distributions for tomogram 7 before and after supervised adaptation. The base model produces very few correct matches and misses most particles, whereas the adapted model recovers almost the complete ground-truth field with far fewer misses.

Table 4.2 also exposes the main failure mode that remains after adaptation. Performance is strong and consistent in four tomograms, but tomogram 8 is visibly harder. Figure 4.2 shows the mechanism on a representative validation tomogram: adaptation changes the model from a detector that misses almost everything into one that covers the particle field with only a small number of residual FP and FN. After adaptation, the main remaining issue is reduced selectivity under difficult noise conditions. This is why post-adaptation robustness must be evaluated at the tomogram level rather than only through a global mean.

4.3. Most performance is recovered with few tomograms and multi-prompt inference reduces false positives

Having established that supervised adaptation is necessary for the synthetic thyroglobulin target, we next want to know how much annotated data is needed to recover most of the attainable performance, and whether prompt aggregation provides a measurable advantage over a single prompt. If the model adapts efficiently, the learning curve should rise sharply at small sample sizes and then saturate. If prompt aggregation stabilizes the conditioning signal, its advantage should appear mainly through FP reduction and therefore through precision.

For this, the experiment reuses the same validation set as EXP2 and trains seven checkpoints with $1, 2, 4, 8, 12, 16$, and 20 training tomograms. The adaptive-epoch policy ensures that small subsets receive longer optimization and are not penalized by construction.

The experiment also introduces the first fully explicit prompt-selection strategy of the work. The single-prompt branch uses a quality score based on three interpretable terms:

- A Fourier mid-band to high-band energy ratio, which favors structured signal over high-frequency noise.
- A center-to-border ratio, which favors crops whose main mass is centered.
- A DC penalty, which discourages crops dominated by offset or gradient artifacts.

The multi-prompt branch selects ten diverse crops by variance ranking, encodes them with TomoTwin and represents the class through both the first embedding and the mean embedding. The resulting prompt statistics show that the cosine similarity between the single and multi representations is 0.9629, with mean embedding-dimension variance 0.010789. This is important because it shows that the two prompt representations correspond to closely related instances of the same particle class; the expected gain is therefore robustness, not a change of class semantics.

	Single prompt				*Multi-prompt* $n = 10$			
Training tomograms	F_1	TP	FP	FN	F_1	TP	FP	FN
1	0.691	468	153	154	0.647	384	60	238
2	0.728	488	124	134	0.717	440	54	182
4	0.800	604	285	18	0.875	578	118	44
8	0.860	609	186	13	0.906	595	95	27
12	0.862	611	190	11	**0.919**	590	70	32
16	**0.888**	612	149	10	0.911	590	83	32
20	0.849	611	220	11	0.894	595	118	27

Table 4.3: EXP3 grouped confusion-count summary. F_1 is reported as the mean over five validation tomograms, whereas TP, FP and FN are summed over the same split.

In the repository, the EXP3 notebooks contain the physical prompt-quality score, build single and multi-prompt embeddings, evaluate the training increments and compute the fixed matching metrics. The appendix keeps only those algorithmic fragments because they are the parts that affect the scientific comparison.

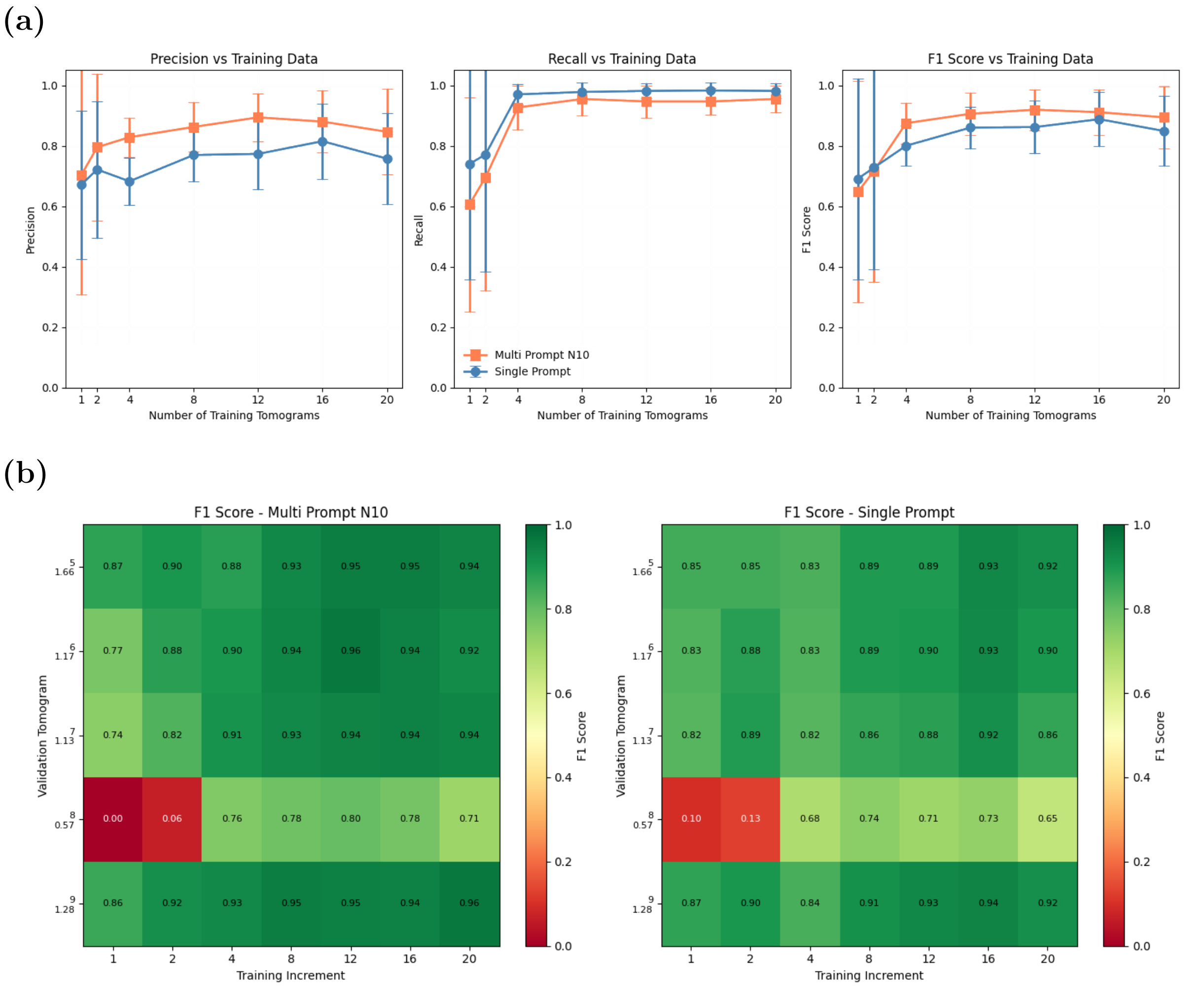


Figure 4.3: EXP3 data efficiency and multi-prompt inference. (a) Mean precision, recall and F_1 as a function of training-set size. Each point is the mean over the five fixed validation tomograms; the vertical interval is computed as the corresponding sample standard deviation across those five tomogram-level measurements. Most of the gain appears early and the multi-prompt branch overtakes the single-prompt branch from 4 tomograms onward, with the best observed result at 12 tomograms. **(b)** Validation-tomogram heatmaps for F_1, showing that the improvement is broadly shared across the fixed split while tomogram 8 remains the persistent hard case for both prompt strategies.

Figure 4.3 makes the EXP3 story visible at two complementary levels. Panel (a) shows the global learning curves: both strategies improve rapidly with few tomograms, the gain after 4 tomograms is already substantial and the multi-prompt branch becomes clearly superior from that point onward. The best configuration is reached at 12 tomograms with multi-prompt inference, obtaining $F_1 = 0.919$, after which the curve saturates.

Panel (b) adds the tomogram-level view. Most validation tomograms stabilize at high F_1 once 4 to 8 training tomograms are available, but tomogram 8 remains consistently harder than the other four volumes. It shows that the aggregate curve reflects a consistent learning trend across the validation set, while the residual weakness is concentrated in the same difficult tomogram. Table 4.3 then explains why multi-prompt wins operationally.

From 4 tomograms onward it usually gives up a small number of TP in exchange for a much larger FP reduction, which is exactly the tradeoff that improves precision and raises the final F_1.

If annotation time is limited, the results indicate that 4 tomograms already define a strong operating point. More importantly, the multi-prompt strategy reaches better results with less training data: with only 4 tomograms it already surpasses the single-prompt branch at 8 tomograms and it stays ahead thereafter. Under the current protocol, the strongest observed configuration is 12 tomograms with multi-prompt inference. Figure 4.3 demonstrates that this pattern is supported by both the mean curve and the tomogram-level heatmap. The advantage is not an artifact of one unusually easy validation volume. All tomograms except the 8th climb into a high-performance regime quickly, while tomogram 8 remains the systematic outlier under both prompt strategies.

The reason why multi-prompt is preferable in this experiment is not that it recovers more particles at all costs, but that it suppresses FP much more effectively. At the 12-tomogram setting, the single-prompt model yields 190 FP, while the multi-prompt model reduces them to 70, with only a moderate loss in TP. A detector with slightly lower recall but far fewer FP may be operationally preferable because it reduces manual curation and prevents contamination of downstream subtomogram processing. Methodologically, this behavior shows that prompt selection cannot remain an informal choice once the detector is already strong. The implementation of prompt ranking, prompt aggregation and tomogram-level failure analysis becomes part of the model-selection problem itself.

4.4. Prompt performance is not a simple function of the studied acquisition parameters

The last question is whether an adapted checkpoint remains stable when the prompt particle changes. If average performance across several prompts is high while a few individual prompts fail, the remaining weakness is prompt representativeness rather than a globally weak detector. The expected failure mode is recall collapse, not an explosion of false detections.

For this, EXP4 first evaluates 10 rotation-diverse training prompts using the fixed checkpoint adapted using 16 tomograms with multi-prompt aggregation, and then expands the analysis to a larger prompt population, always using only training tomograms to avoid validation leakage. Candidate selection is quality-aware and rotation-aware: from 2705 training particles, the notebook keeps 2165 valid coordinates, removes 1126 border-near particles, scores 1039 crop-safe candidates and retains 364 high-quality candidates before selecting the final 10 prompts. These prompts come from four training tomograms and remain well separated, with pairwise angular distances from 105.8° to 179.9°. The large-population notebook applies the same idea in $SO(3)/C_2$, respecting thyroglobulin C2 symmetry, and analyzes the first 125 prompts from a 200-candidate run.

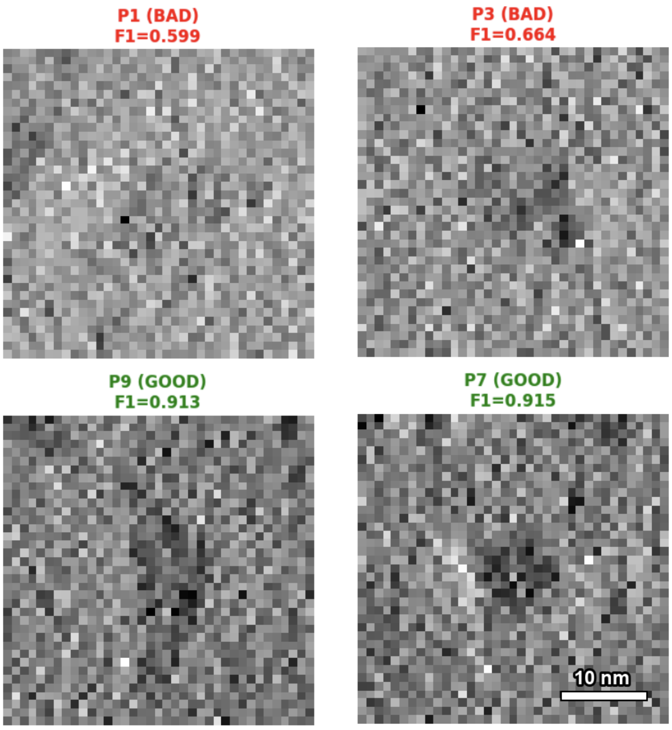


Figure 4.4: EXP4 prompt crop examples. Weak prompts show lower contrast and less clear particle structure than the strongest prompts, so poor downstream behavior is not explained by obvious crop noise alone.

The first part asks whether the selected prompts behave equivalently after adaptation. Figure 4.4 shows that weak prompts are not simply the noisiest crops. Compared with the stronger prompts, they have visibly lower contrast and less clear particle structure.

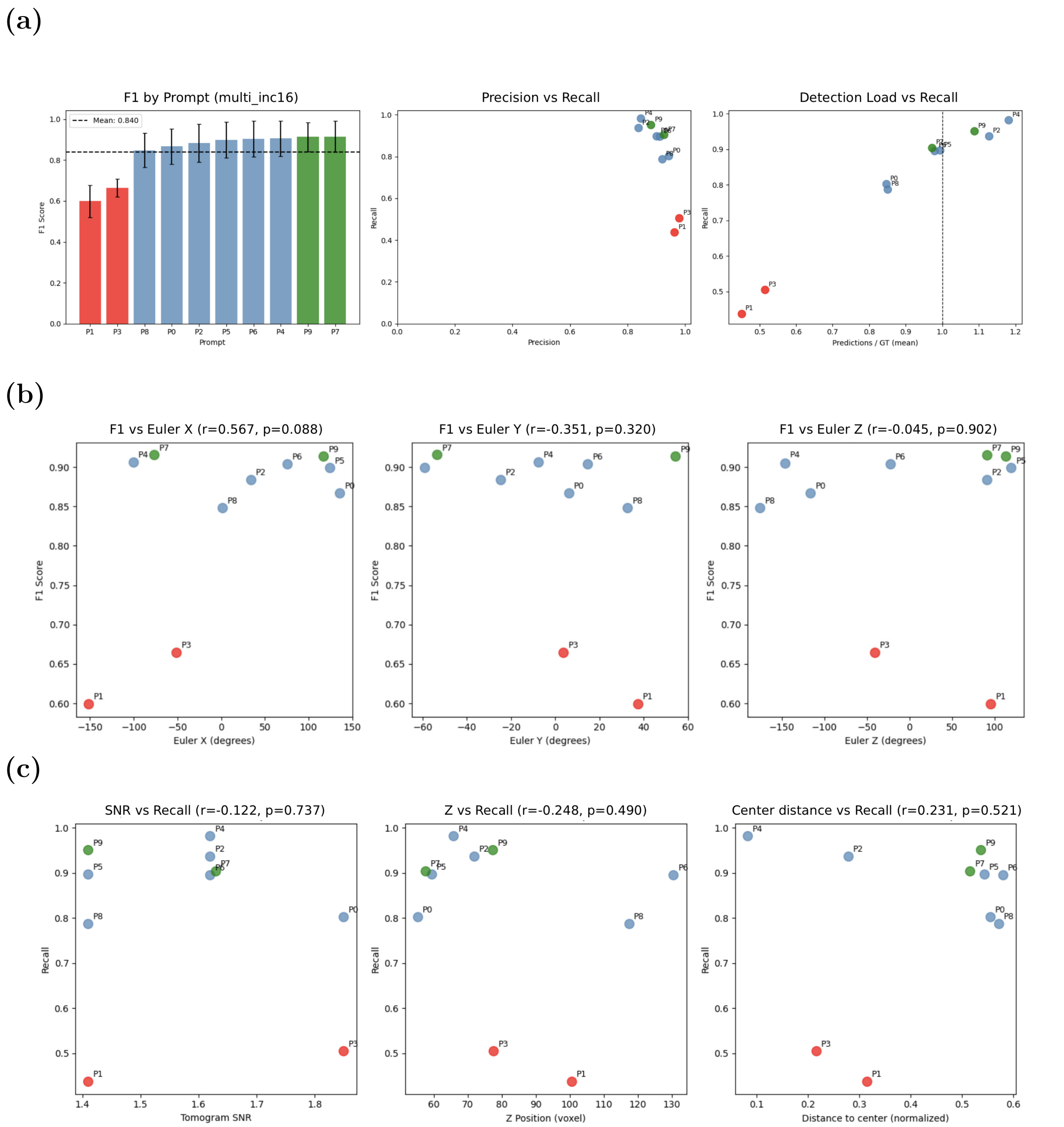


Figure 4.5: EXP4 prompt metric and rotation diagnostics for the `multi_inc16` checkpoint. **(a)** Prompt-level metrics identify P1 and P3 as the main weak prompts. **(b)** Euler-angle views of the prompt set. **(c)** Metric comparison across the prompt set.

Mean single-prompt performance remains high ($F_1 = 0.840 \pm 0.130$, precision $= 0.912$, recall $= 0.809$), but Figure 4.5 shows the same weak-prompt pattern across metric views: P7 and P9 reach $F_1 = 0.915/0.913$, whereas P1 and P3 fall to $F_1 = 0.599/0.664$. Their failure is mainly under-activation, with high precision (0.964/0.980) but low recall (0.437/0.504). Tomogram 8 also remains the hardest validation volume, with mean $F_1 = 0.701$. The quality score, Euler-angle views and prediction counts do not support rotation alone or another single-factor explanation such as crop noise, repeated view sampling or one acquisition descriptor. Max aggregation partly buffers isolated weak prompts, reaching recall 0.9818 but lower precision 0.7067, for $F_1 = 0.8169$; removing P1 and P3 does not change that aggregate metric.

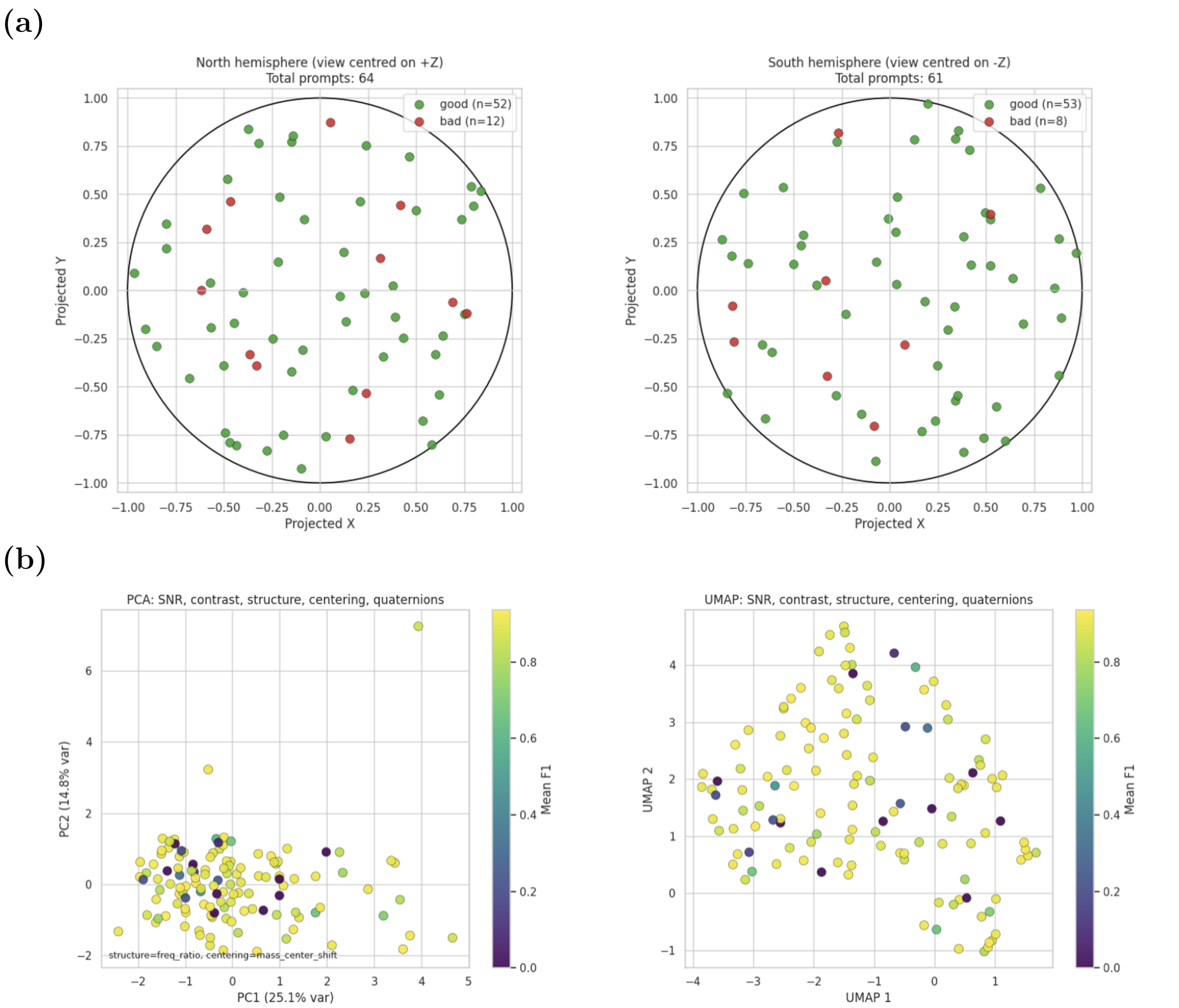


Figure 4.6: EXP4 large prompt-population diagnostics. **(a)** The local-Z orientation map places good and bad prompts in both hemispheres, ruling out a single global orientation side as the cause of failure. **(b)** PCA and UMAP views over SNR, contrast, structure, centering and quaternion descriptors indicate broad descriptor structure, but not a single clean cluster that explains all low-F_1 prompts.

The expanded 125-prompt study confirms that prompt variability is not a small-sample artifact: $F_1 = 0.780 \pm 0.283$, precision $= 0.824$ and recall $= 0.783$, with tomogram 8 again the hardest validation volume ($F_1 = 0.659$). Source SNR, source Z and distance to the tomogram center remain weak, non-significant recall predictors ($|r| \leq 0.248$, $p \geq 0.490$), and Figure 4.6 rules out a single geometric explanation: good and bad prompts appear in both hemispheres, and PCA/UMAP do not isolate all low-F_1 prompts. The remaining signature is mixed, with bad prompts tending toward lower prediction load, lower contrast, lower center distance and larger mass-center shift. The follow-up notebook ranks 30 similar candidates and selects 10 for future reproduction testing, but this work treats that as exploratory. The final TomoTwin input should therefore be understood as an average of several clean and diverse prompt embeddings, designed to reduce dependence on any single representative particle.

Chapter 5

Conclusions

This work set out to evaluate whether a promptable DL detector can act as a practical step toward universal particle localization in cryo-ET. The experiments show that ProPicker is useful when the target particle is adapted with supervised data and prompt construction is controlled. It is not yet universal in the stronger sense of transferring reliably to every new particle or acquisition condition from a single prompt and the base checkpoint alone.

5.1. Summary findings

The first finding is that prompt-only transfer is informative but incomplete. On EMPIAR-10988, the base model detects ribosome signal but with low recall. Threshold calibration slightly improves the operating point, while fine-tuning produces the real change and raises the validation F_1 from 0.3074 to 0.5714. This validates the pipeline and shows that the model contains useful transferable structure, but it also makes clear that the base model should not be treated as a final detector on real data without adaptation.

The second finding is that transfer to a new particle target and dataset remains a hard limit. On the PolNet thyroglobulin dataset, the pretrained detector collapses to $F_1 \approx 0.008$, while supervised adaptation reaches a mean F_1 of 0.891 on the fixed validation split. This is the clearest boundary found in this work: universality cannot be claimed from promptability alone. When the particle class, tomogram statistics or reconstruction conditions change, supervised adaptation is the step that makes the method operational.

The third finding is important from an annotation-cost perspective. The incremental study shows that strong performance appears with relatively little training data. Four tomograms already define a useful operating point, and the best observed configuration is obtained with 12 training tomograms and multi-prompt inference, reaching $F_1 = 0.919$. The multi-prompt approach mainly reduces FP, which is exactly the type of improvement that makes later subtomogram extraction and manual inspection easier.

The fourth finding is that the main unresolved problem after adaptation is prompt robustness. EXP4 shows that a strong checkpoint can still fail for particular prompts, mostly through recall collapse rather than through excessive FP. In the 10-prompt study, two are weak despite being selected by the same quality and diversity procedure as the stronger prompts. In the larger prompt population, the same instability remains visible, and neither source SNR, Z position, distance to tomogram center nor a single orientation rule explains it on its own. The interpretation is that prompt quality depends on the joint effect of crop content, anisotropic acquisition, thyroglobulin symmetry and latent-space representativeness. This makes prompt selection part of the detector design.

5.2. Limitations

These conclusions should be read within the scope of the experiments. The real-data component is a controlled proof of concept rather than a full EMPIAR-scale evaluation. The completed synthetic study is single-class, even though the repository is structured to support broader extensions. Some post-processing thresholds are inherited from the reference workflow and are kept fixed for comparability rather than rederived exhaustively in every setting. Finally, prompt selection is itself a source of variance, so the reported metrics must always be interpreted together with the prompt protocol used to obtain them. The EXP4 follow-up candidate search is also exploratory: it identifies prompts that should be tested next, but it is not a substitute for a completed reproduction run.

5.3. Future work

The most immediate future direction is automatic prompt screening. A deployment-oriented workflow should rank candidates by structural quality, centering, diversity and representativeness before inference, and the EXP4 candidate set provides a concrete starting point for testing whether weak-prompt behavior can be reproduced prospectively. A second direction is broader validation on additional real datasets, more particle families and multiclass settings. A third direction is to make adaptation cheaper through lightweight fine-tuning, active learning or human-in-the-loop refinement. A fourth direction is to incorporate cryo-ET physics more directly, for example through missing-wedge-aware augmentation and symmetry-aware evaluation in spaces such as $SO(3)/C_2$.

The final conclusion is therefore deliberately qualified. ProPicker is not yet a universal cryo-ET detector in the strict sense, but it is already a strong adaptable detector. With consistent preprocessing, supervised adaptation, multi-prompt conditioning and explicit failure analysis, it provides a credible foundation for building more reliable universal detectors for macromolecule localization.

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Appendix A

Hardware Specifications and Software Stack

A.1. AI usage

Artificial intelligence tools were used during this work as support instruments for code assistance and writing checks. In the software part, AI support helped inspect scripts, identify implementation issues, propose small code improvements and review command-line workflows. In the writing part, AI support was used to check clarity, consistency, grammar and structure in selected fragments of the document. The experimental design, execution of the experiments, interpretation of the results and final scientific decisions remained under the responsibility of the author.

A.2. Documented experimental hardware

Cryo-ET model adaptation and tomogram inference are computationally demanding tasks that depend on graphics processing unit (GPU) capacity, memory availability and storage throughput. For this reason, the experimental pipeline was executed on a dedicated workstation whose main specifications are summarized in Table A.1.

Component	Documented value
OS	Ubuntu 24.04.3 LTS (kernel 6.17.0-20-generic)
CUDA version	13.0
Main training GPU	NVIDIA GeForce RTX 4090 (24 GB VRAM)
CPU	13th Gen Intel(R) Core(TM) i7-13700KF
RAM	$\approx$ 135 GB

Table A.1: Experimental hardware information of the AIKE laboratory workstation.

A.3. Local authoring and validation environment

Part of the writing and repository inspection for this document was carried out on the local system shown in Table A.2. This machine should not be confused with the documented experimental training workstation.

Component	Value
Computer	MacBook Pro (Mac14,10)
CPU	Apple M2 Pro, 12 CPU cores (8 performance + 4 efficiency)
Memory	16 GB unified memory
Integrated GPU	Apple M2 Pro, 19 GPU cores

Table A.2: Local authoring environment used for the present manuscript work.

A.4. Software, languages and technologies

The project relies on standard technologies widely used in scientific computing summarized in Table A.3, chosen for their mature documentation, compatibility with cryo-ET data formats and straightforward integration with the ProPicker workflow.

Category	Tool or technology	Role in the project
Programming	Python	Main implementation language for preprocessing, adaptation, inference and analysis.
Manuscript	LaTeX	Preparation of this document.
Data formats	JSON, CSV, MRC	Prompt embeddings, coordinate tables and tomogram/localization-map storage.
Experiments	Jupyter Notebook	Exploratory preprocessing, metric analysis, prompt inspection and plotting.
Core detector	ProPicker	Promptable 3D segmentation model used as the central detector platform.
Training model	DeepETPicker	Generation of preprocessing and training configs and command-line training/inference execution.
Prompt encoder	TomoTwin	Extraction of prompt embeddings from subtomograms of 37^3 voxels.
Main libraries	PyTorch, NumPy and Pandas	Checkpoint loading, inference-time handling of localization maps, numerical operations and table manipulation.
Volume input/output	`mrcfile` and toolchain MRC utilities	Reading and writing tomograms, labels and localization maps.
Visualization	ParaView	Inspection of tomograms and exported localization maps in volumetric form.
Environment	Conda environment	Execution of the ProPicker/DeepETPicker training and inference stack.
Version control	Git	Repository tracking.

Table A.3: Software, languages and technologies used.

Appendix B

Repository Structure and Implementation Details

The source code and implementation for this project are available in the public GitHub repository: `https://github.com/C-HernanG/cryoet-particle-picking`.

B.1. Current repository scaffolding

The repository summarized in Table B.1 was designed for scalability to facilitate the seamless integration of future experiments.

Component	Main role	Practical guideline
`paths.py`	Central path registry	Update this file first when moving the project to a new machine. All experiment scripts depend on it.
`config.py`	Shared experimental constants	Keep common splits and hyperparameters here so that EXP2, EXP3 and EXP4 remain directly comparable.
`data`	Input datasets	Store EMPIAR and PolNet tomograms, annotations and other raw inputs here.
`experiments`	Experiment workspace	Store the specific experiment folders, notebooks and scripts here.
`results`	Derived artifacts	Prompts, coordinates, checkpoints, inference outputs and analysis tables should be written here rather than into the source folders.
`docs`	Documentation and reports	This document, interim reports and future experiment specifications are maintained here and versioned together with the code.
`models`	Base checkpoints	Store all the checkpoints used for experiments.
`tools`	Dependencies	Keep the cloned ProPicker here.

Table B.1: Repository scaffolding used by the investigation workflow.

B.2. Minimal reproducibility checklist

For a reader who wants to rerun or extend the investigation, the minimum checklist is the following:

1. Configure `paths.py` for the local machine.
2. Place the ProPicker and TomoTwin checkpoints in the `models` folder and clone ProPicker into the `tools` folder.
3. Install the Python environment used by DeepETPicker and ProPicker.
4. Populate `data` directory with the EMPIAR or PolNet datasets.
5. Execute the notebook that generates prompts and coordinates for the experiment.
6. Run the training or inference script corresponding to that experiment.

The role of this checklist is to make the repository intelligible as a research instrument and to document the scaffolding assumptions on which the results of the work depend.

B.3. Condensed Algorithm Listings

The notebooks used in the experiments contain a large amount of supporting code for plotting, file management and interactive inspection. For the appendix, the most useful compromise is to keep only the core algorithmic fragments that define how prompts are scored, selected, aggregated and evaluated. The training and inference wrappers are described in the corresponding experiment sections, because they mainly set paths, generate DeepETPicker configuration files and call ProPicker entry points.

B.3.1. EXP3 prompt scoring and single-prompt selection

The core prompt-selection logic used in `exp3_ppicker_limits.ipynb` combines three physically motivated features: a mid-frequency versus high-frequency ratio, a center-versus-border energy ratio and a penalty for global offset. This is the physical prompt-quality score used for single-prompt selection in EXP3. In the full notebook, helper functions perform robust normalization and optional embedding-consensus refinement; Listing B.1 keeps only the essential decision rule.

```
def compute_quality_score(subtomo, border=4, centre_frac=0.45,
                          k_low=0.08, k_high=0.30):
    vol = robust_norm(to_numpy(subtomo))
    freq_ratio = freq_structure_ratio(vol, k_low=k_low, k_high=k_high)
    centre_ratio = center_border_ratio(
        vol, border=border, centre_frac=centre_frac
    )
    dc_penalty = abs(vol.mean()) / (vol.std() + 1e-6)

    score = np.log(freq_ratio) + 0.9 * np.log(centre_ratio) - 0.6 * dc_penalty
    return score, {
        "freq_ratio": freq_ratio,
        "centre_ratio": centre_ratio,
```

```python
        "dc_penalty": dc_penalty,
    }


def select_best_prompt(subtomos, prompt_encoder=None, top_k=8):
    base_scores = np.array([compute_quality_score(v)[0] for v in subtomos])
    candidate_idx = np.argsort(base_scores)[::-1][:top_k]
    final_scores = base_scores.copy()

    if prompt_encoder is not None and len(candidate_idx) >= 2:
        E = np.stack([prompt_encoder(subtomos[i]).reshape(-1)
                      for i in candidate_idx])
        S = cosine_similarity_matrix(E)
        consensus = (S.sum(axis=1) - 1.0) / (S.shape[0] - 1)
        for j, i in enumerate(candidate_idx):
            final_scores[i] += 1.2 * consensus[j]

    best_idx = int(np.argmax(final_scores))
    return best_idx, subtomos[best_idx]
```
Listing B.1: Condensed EXP3 single-prompt selector based on subtomogram quality.

This selector reflects an important methodological shift in the work: once detector performance becomes competitive, prompt choice itself becomes part of model selection rather than a manual convenience step.

B.3.2. EXP3 multi-prompt aggregation

The second important idea introduced in `exp3_ppicker_limits_inference.ipynb` is that the operational prompt does not have to be a single subtomogram embedding. In the experiment, several prompt embeddings are computed from selected particles and their arithmetic mean is used as the «multi-prompt» representation. This is the embedding-construction step used before evaluating the training increments. Listing B.2 shows the compact form of that logic.

```python
def build_prompt_embeddings(subtomos, prompt_encoder, n_instances=10):
    embeddings = torch.stack([
        prompt_encoder(subtomo).reshape(-1)
        for subtomo in subtomos[:n_instances]
    ])

    single_prompt = embeddings[0]
    multi_prompt = embeddings.mean(dim=0)

    return {
        "single_prompt": single_prompt,
        "multi_prompt": multi_prompt,
        "all_embeddings": embeddings,
    }
```
Listing B.2: Condensed EXP3 multi-prompt aggregation. In the notebook, the input subtomograms are already pre-selected by the prompt-scoring stage.

In the actual experiment, this averaged embedding produced the best observed operating mode because it reduced FP more strongly than it reduced recall.

B.3.3. EXP4 Step 2: quality-filtered rotation-diverse prompt selection

Step 2 of `exp4_large_prompt_population_inference.ipynb` extends prompt selection by adding quaternion distances and a symmetry-aware interpretation for thyroglobulin. This is the large-population construction used in EXP4. The selector first filters out unsafe border candidates, then keeps only high-quality subtomograms and finally builds a prompt population that is as diverse as possible in orientation space while still preferring better-quality candidates. Listing B.3 gives the condensed form of the notebook logic.

```
def quaternion_angular_distance_c2(q1, q2, c2_rotation):
    r1 = Rotation.from_quat(normalize_quaternion(q1))
    r2 = Rotation.from_quat(normalize_quaternion(q2))
    return min(
        np.degrees((r1.inv() * r2).magnitude()),
        np.degrees((r1.inv() * (r2 * c2_rotation)).magnitude()),
    )


def select_diverse_rotations_with_quality(
    df, tomo_cache, n_samples, min_angular_dist=30.0,
    quality_percentile=65.0, max_per_tomo=None
):
    df = filter_border_safe_candidates(df, tomo_cache, min_border_margin=50)
    df["quality_score"] = [
        compute_quality_score(extract_subtomo(row, tomo_cache))[0]
        for _, row in df.iterrows()
    ]

    threshold = np.nanpercentile(df["quality_score"], quality_percentile)
    high_quality = df[df["quality_score"] >= threshold].copy()
    if len(high_quality) < n_samples:
        threshold = np.sort(df["quality_score"].to_numpy())[-n_samples]
        high_quality = df[df["quality_score"] >= threshold].copy()
    df = high_quality

    quats = df[["Q1", "Q2", "Q3", "Q4"]].to_numpy()
    scores = df["quality_score"].to_numpy()
    tomo_names = df["tomo_name"].to_numpy()

    selected = [int(np.argmax(scores))]
    tomo_counts = {tomo_names[selected[0]]: 1}

    while len(selected) < min(n_samples, len(df)):
        best_idx, best_min_dist, best_quality = -1, -1.0, -np.inf
        for i in range(len(df)):
            if i in selected:
                continue
            if (
                max_per_tomo is not None
                and tomo_counts.get(tomo_names[i], 0) >= max_per_tomo
            ):
                continue

            min_dist = min(
                quaternion_angular_distance_c2(
```

```
                    quats[i], quats[j], THYROGLOBULIN_C2_ROTATION
                )
                for j in selected
            )

            if min_dist >= min_angular_dist and (
                min_dist > best_min_dist or
                (min_dist == best_min_dist and scores[i] > best_quality)
            ):
                best_idx, best_min_dist, best_quality = i, min_dist, scores[i]

        if best_idx == -1:
            for i in range(len(df)):
                if i in selected:
                    continue
                if (
                    max_per_tomo is not None
                    and tomo_counts.get(tomo_names[i], 0) >= max_per_tomo
                ):
                    continue
                min_dist = min(
                    quaternion_angular_distance_c2(
                        quats[i], quats[j], THYROGLOBULIN_C2_ROTATION
                    )
                    for j in selected
                )
                if min_dist > best_min_dist or (
                    min_dist == best_min_dist and scores[i] > best_quality
                ):
                    best_idx, best_min_dist, best_quality = i, min_dist, scores[i]
            if best_idx == -1:
                break

        selected.append(best_idx)
        tomo_counts[tomo_names[best_idx]] = (
            tomo_counts.get(tomo_names[best_idx], 0) + 1
        )

    return df.iloc[selected].copy()
```

Listing B.3: Condensed EXP4 selector for quality-filtered and rotation-diverse prompts.

The key research idea combines quaternion-based rotation handling with symmetry-aware prompt selection. This is precisely what made it possible to separate prompt quality effects from naive «bad rotation» explanations in the final analysis.

B.3.4. Fixed greedy evaluation rule used in the synthetic experiments

Both `exp3_ppicker_limits_inference.ipynb` and `exp4_large_prompt_population_inference.ipynb` keep a fixed center-matching rule between predicted coordinates and ground truth inside each analysis. EXP3 uses 18.5 voxels, equal to half the prompt size of 37^3 voxels. EXP4 uses 15.0 voxels in the large prompt-population analysis to make prompt-level failures stricter. Listing B.4 shows the compact implementation. The matching is greedy and one-to-one: once a ground truth point has been matched, it cannot be reused by another prediction.

```python
def compute_metrics(pred_coords, gt_coords, distance_threshold):
    if len(pred_coords) == 0:
        return {"precision": 0.0, "recall": 0.0, "f1": 0.0,
                "tp": 0, "fp": 0, "fn": len(gt_coords)}
    if len(gt_coords) == 0:
        return {"precision": 0.0, "recall": 0.0, "f1": 0.0,
                "tp": 0, "fp": len(pred_coords), "fn": 0}

    distances = cdist(pred_coords, gt_coords)
    matched_gt = set()
    tp = 0

    for i in range(len(pred_coords)):
        best_j, best_d = -1, np.inf
        for j in range(len(gt_coords)):
            if j not in matched_gt and distances[i, j] < best_d:
                best_j, best_d = j, distances[i, j]

        if best_j != -1 and best_d <= distance_threshold:
            matched_gt.add(best_j)
            tp += 1

    fp = len(pred_coords) - tp
    fn = len(gt_coords) - tp
    precision = tp / (tp + fp) if tp + fp else 0.0
    recall = tp / (tp + fn) if tp + fn else 0.0
    f1 = (2 * precision * recall / (precision + recall)
          if precision + recall else 0.0)

    return {"precision": precision, "recall": recall, "f1": f1,
            "tp": tp, "fp": fp, "fn": fn}
```

Listing B.4: Condensed greedy one-to-one coordinate matching used to compute precision, recall and F_1.

Keeping this rule exactly the same across all synthetic experiments is scientifically important. It ensures the dataset stays constant even when we change the prompting strategy, model checkpoint or number of training tomograms.